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Gates Millennium Scholars (GMS) Survey Data Cohort 5, United States, 2004-2009

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Final Reports and Study Methodology Materials

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The Fifth Cohort of Gates Millennium Scholars One Year After High School

A Comparative Analysis of the Background Characteristics, Early Outcomes, and Experiences of the Entering Freshman Class of 2004

A Report Published by
The National Opinion Research Center (NORC)

Prepared for
The Bill & Melinda Gates Foundation

April 2007

The Gates Millennium Scholars program (GMS), sponsored by the Bill & Melinda Gates Foundation, is designed to broaden opportunity and access to higher education for low-income, high-achieving African Americans, American Indians/Alaska Natives, Asian/Pacific Islanders, and Hispanic Americans.

The National Opinion Research Center (NORC) at the University of Chicago conducts the GMS Tracking and Longitudinal Study, which collects data on the outcomes and experiences of select cohorts of applicants to the GMS program.

This and other selected reports are available by request from the National Opinion Research Center, 1155 E. 60th Street, Chicago, Illinois 60637, as well as on the project's website at http://www.norc.org/gatesscholars/.

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FOREWORD

This report documents preliminary findings from the baseline survey of the entering freshman class of the fifth cohort of Gates Millennium Scholars. It is intended to serve as a reference source for information on the background characteristics, early outcomes, and experiences of these Scholars and a comparison sample of non-recipients. In a series of exhibits, it presents survey estimates, along with their standard errors, for the Scholar and non-recipient populations as a whole, and by gender and minority group within each population. Along with a description of the information in each exhibit, it discusses the results of an exploratory analysis—highlighting patterns found within and across the Scholar and non-recipient populations.

The report focuses on describing the Scholar and non-recipient populations along with their similarities and differences. It does not attempt to separate the effects of the scholarship program from other factors, including differences between the background characteristics of the Scholar and non-recipient populations.

The report begins by summarizing some of the key findings of the exploratory analysis. The main body of the report discusses those and other findings in detail after providing information on the scholarship program and the methodology of the study. The appendices at the end of the report discuss the technical aspects of the survey and the exploratory analysis in detail.

Please note that the findings discussed in this report do not necessarily reflect the views of the Bill & Melinda Gates Foundation or its staff.

Acknowledgements

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We especially want to thank the NORC staff members who helped in various roles in the preparation of the report. Bronwyn Nichols Lodato coordinated the staff for the report; Isabel Guzman-Barron assisted in the formatting and placement of the exhibits; Chris Bergstresser programmed the production of the report tables; Emily Weaver reviewed the content of the report, including the numerical information and statistical computations; Jill Connelly and Lynette Bertsche did the final proofreading and copy editing of the report; Imelda Demus was responsible for the graphic design and formatting of the report; and Patricia Cloud was responsible for the publication and cataloging of the report.

Victor Kuo, Program Officer in Education, and Margot Tyler, Senior Program Officer (the Bill & Melinda Gates Foundation), provided helpful comments on an earlier draft of this report.

Any errors that may remain are the responsibility of the authors.

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Selected Key Findings

Background Characteristics

- While Scholars were selected without regard to gender, there are more than twice as many women as men among this population. Among the non-recipients as well, there are also more than twice as many women as men.
- Both Scholars and non-recipients report origins in over 30 different countries and islands from around the world with substantial numbers from Mexico, Vietnam, and China.
- Both American Indian/Alaska Native Scholars and non-recipients are affiliated with over 40 different state and federally recognized American Indian and Alaska Native tribes.
- In nearly all of the country's nine Census Divisions, there is a group of outstanding high school seniors in each minority group who applied to the GMS program for aid and made it through the selection process to become Scholars. There are also non-recipients from each minority group in nearly every Census Division.
- The parents of Hispanic American Scholars show overall lower levels of educational attainment compared to the parents of Scholars from other minority groups. Also, the Scholars' parents completed high school at lower rates than their non-recipient counterparts.

High School Experiences

- Most Scholars were in high school classes with sizable shares of students from racial/ethnic backgrounds other than their own. The non-recipients were in classes that were more racially diverse than the Scholars' classes.
- About three out of every four Scholars took at least one advanced placement exam while in high school. More than one-third took four or more. The non-recipients as a whole have taken advanced placement exams at lower rates than Scholars; non-recipients were both less likely to have taken at least one advanced placement exam and less likely to have taken four or more exams.
- Most Scholars and non-recipients attended public high schools. However, about 17 percent of American Indian/Alaska Native Scholars attended a private high school, significantly more than the other three minority groups.
- The overwhelming majority of Scholars took at least three or more years of math and three or more years of science while in high school. Most took four or more. The Scholars were more likely to have taken four years of math and science in high school than were the non-recipients.

Educational Aspirations and College Enrollment

- Almost all Scholars aspired to earn at least a bachelor's degree and most also hoped to earn at least a master's degree. Nearly two-thirds aspired to attain a professional or doctoral degree. As a whole, non-recipients were less likely than Scholars to believe they would complete an advanced degree. For every minority group, the Scholars were more likely than their non-recipient counterparts to believe they would earn at least a master's or a first professional or doctoral degree.

- Nearly all Scholars and non-recipients were currently enrolled in college at the time of the baseline survey. For just over half of the Scholars, their current institution (or institution of last enrollment if not currently enrolled) was a public institution, and for almost all of the Scholars their institution offered at least a bachelor's degree. In comparison, somewhat more non-recipients, about two-thirds, were either currently or last enrolled in public institutions, and somewhat fewer, close to 93 percent, were enrolled at institutions offering at least a bachelor's degree.

- For most Scholars their current or last institution of enrollment offered at least a bachelor's degree. However, American Indian/Alaska Native Scholars were less likely than Scholars from other minority groups to enroll in institutions offering at least a bachelor's degree.

Working for Pay

- Working for pay while in college was common among both Scholars and non-recipients, but as a whole Scholars were less likely than non-recipients to work for pay. About half of the Scholars were working compared to close to two-thirds of the non-recipients.

- With regard to their reasons for working, Scholars were primarily working to meet living expenses or to make money. Non-recipients were primarily working as part of a work-study program/to pay for school or to meet living expenses. Compared to non-recipients, Scholars were more likely to be working for the learning experience, to support their family, or as part of their career path or internship. Compared to Scholars, non-recipients were more likely to be working to pay back loans or debt or as part of a work-study program/to pay for school.

Other Sources of Financial Support

- Scholars were far less likely than non-recipients to have taken out a loan. About one-fifth of Scholars took out loans during the current academic year, while almost three-fifths of non-recipients did so. When they took out loans, non-recipients were more likely than Scholars to borrow money to pay for tuition and for room and board; Scholars were more likely than non-recipients to borrow money to pay for "other" expenses. Even so, both Scholars and non-recipients cited paying for tuition, room and board, and books and other educational supplies as their top three reasons for taking out loans.

- Scholars were less likely than non-recipients to receive assistance from their parents in financing college. About one-third of Scholars received assistance from their parents. In comparison, more than half of non-recipients received help from their parents.

A Comparative Analysis of the Background Characteristics, Early Outcomes, and Experiences of the Entering Freshman Class of 2004

Introduction

In 1999, the Bill & Melinda Gates Foundation established The Gates Millennium Scholarship program (GMS), a twenty-year initiative to provide high-achieving, low-income students of color with an opportunity to attend and complete college and, for those wishing to continue their studies in a discipline where their minority group is underrepresented, a chance to attend and complete graduate school. In 2000, the first award year of the program, the United Negro College Fund (UNCF), as the program's administrator, awarded over 4,000 grants to outstanding minority students who were entering college or continuing their undergraduate or graduate work in the 2000-2001 academic year. Every year since then, UNCF has awarded close to 1,000 additional scholarships to entering freshmen to fund their studies as they begin and progress through college. By the year 2016, the program will have helped close to 20,000 minority students with significant financial need attend the colleges and universities of their choice and prepare for successful careers.

To assess the impact of the scholarship program and, more generally, to better understand the key practices, experiences, and attributes that help prepare minority students for college and contribute to their sustained success, the Bill & Melinda Gates Foundation asked the National Opinion Research Center at the University of Chicago (NORC) to follow selected cohorts of GMS recipients (Scholars) and comparison samples of non-recipients as they enter and progress through college and beyond. Studies of the inaugural, second, third, and fifth cohorts are currently underway. This report documents some of the preliminary findings from the baseline survey of the entering freshman class of the fifth cohort, Cohort V.

The report is divided into four main parts. Part I discusses the design and methodology of the longitudinal survey and the more technical aspects of the study. Part II describes and compares the background characteristics of the Scholars and non-recipients as a whole, by gender, and by minority group. Part III discusses and compares the college-going rates of the Scholars and non-recipients at the time of the baseline survey and the types of institutions they attended their first year after high school. Finally, Part IV focuses on the early experiences and outcomes of the Scholars and non-recipients as a whole, and by gender and minority group.

PART I:
BACKGROUND AND METHODOLOGY

In the spring of 2005, about a year after they graduated from high school, NORC contacted the fifth cohort of Scholars and a representative sample of 1,333 non-recipients by mail asking them to participate in a web-based survey.

The findings discussed in this report are based on their responses to the survey and on background information supplied by UNCF. In the sections that follow, we present the results from the surveys as estimates of the population and subpopulation values along with a standard error for each value. To compute the population estimates, the respondents to the surveys were case weighted to take into account non-response to the surveys and, in the case of the non-recipients, to adjust for differences in the selection probabilities of the individual sample members and for a small departure from the population/subpopulation totals due to the death of a sample member prior to the onset of the survey. To compute the standard errors of the estimates, NORC relied on the Taylor Series Approximation method as implemented in the SUDAAN statistical software package.[1]

For most survey questions, the categories shown in the exhibits correspond to the response options in the questionnaire. In questions where a response option was not represented in the data (none of the respondents selected the option), the category does not appear in the exhibits. In some questions, the options were recoded into a set of more meaningful or manageable categories. In questions where the Scholars or non-recipients were asked to key in their own responses, their answers were coded into a set of mutually exclusive and exhaustive categories. Throughout the analysis, explicit refusals, don't knows, and not sures were coded as missing values, unless noted otherwise.

The descriptive comparisons discussed in this summary were tested for statistical significance with a two-sided Student's t-test at the 0.05 significance level assuming large sample sizes. For multiple comparisons among the four minority groups on a single measure or between two populations by minority group or by gender on a single measure, a Bonferroni adjustment was applied to take into account the number of comparisons made.

In discussing the results of the analysis, we indicate statistically significant differences between groups by noting that the rate for one group is higher or lower than the rate for another group, one group is more or less likely than the other group, and so on. In discussing the comparisons, we sometimes say that one group "did not differ significantly" from another to indicate that the differences between the groups were not statistically significant. In some cases where no statistically significant differences were found *and* the estimates are close in value, we simply state that the estimates are close in value.

For more information on the technical aspects of the study, refer to Appendices A through F.

[1] It is important to note that the standard errors of the estimates for the American Indians/Alaska Natives tend to be larger than those of the other minority groups due to the small sizes of their Scholar and non-recipient subpopulations and their tendency to participate in the survey at lower rates than the other minority groups. This means that it is more difficult to detect differences between them and the other minority groups when they exist. It is also more difficult to detect differences between the Scholar and non-recipient subpopulations of American Indians/Alaska Natives when they exist.

PART II: A PROFILE OF THE COHORT V SCHOLARS AND NON-RECIPIENTS

2.1 Who Are the Cohort V Scholars?

The Cohort V population of Scholars consists of 1,000 high-achieving minority students with significant financial need who were offered and accepted a GMS award of last-dollar funds for college beginning in the 2004-2005 academic year—their freshman year in college. As in previous cohorts, the population of Scholars is defined as all applicants who applied to the program for aid as entering freshmen, were offered scholarships, and accepted them for their freshman year in college, in this case the 2004-2005 academic year and the fifth award year of the scholarship program. This outstanding group of students was chosen from among thousands of talented youth across the nation on the basis of measures of academic preparation and potential, including high school GPA (3.3 or higher), strength of their high school curriculum, and a student essay, as well as measures of character and leadership potential.[2] Once selected, these students successfully completed the Scholar confirmation/verification process by submitting an official copy of their high school transcripts, proof of their Pell-Grant eligibility, and other supporting documents, at which time they were offered and accepted a GMS scholarship for college.

Exhibit 1 shows that the population of Scholars is made up of students from four minority groups: African Americans, American Indians/Alaska Natives (referred to as American Indians in the exhibits), Asian/Pacific Islanders, and Hispanic Americans. It includes equal numbers of African American and Hispanic American students, with 350 Scholars in each group, and equal numbers of American Indian/Alaska Native and Asian/Pacific Islander students, with 150 in each group. Female Scholars outnumber their male counterparts by about two to one in the Scholar population as a whole.

Exhibit 1. Population Counts for Cohort V Scholars by Minority Group and Gender as Reported on the GMS Application Form

Minority Group	Gender		
	Males	Females	Total
African American	107	243	350
American Indian	42	108	150
Asian/Pacific Islander	45	105	150
Hispanic American	119	231	350
Total Population	313	687	1,000

Source: GMS application form

[2] To qualify for a scholarship, a student must also be a citizen or permanent resident of the United States.

To learn more about the background characteristics of the Scholars, the baseline survey asked the students to answer several questions regarding their geographic, socio-economic, educational, and cultural/racial/ethnic backgrounds. In the exhibits that follow, their responses to those questions are presented as population and subpopulation percents and, in some cases, as averages (means) along with their corresponding standard errors.

In each exhibit, we classified the Scholars into the gender and minority groups shown above in Exhibit 1 according to the information they supplied on the GMS application form. The form asked the Scholars to indicate their membership in *one* of four minority groups: African American, American Indian/Alaska Native, Asian/Pacific Islander, or Hispanic American. Throughout the report, we discuss our findings in terms of those four groups or subpopulations of Scholars.

In the sections that follow, we highlight some of the key findings of the analysis. Please refer to Appendix G for a summary of the key findings by minority group.

2.1.1 Racial and Ethnic Background

The baseline survey asked the Scholars a series of questions regarding their racial and ethnic backgrounds. Exhibits 2 and 3 show the answers they gave as estimates of the population and subpopulation percents.

Race. When given the option of classifying themselves into one or more of five race categories—1) Black or African American, 2) American Indian or Alaska Native, 3) Native Hawaiian or Other Pacific Islander, 4) Asian, and 5) White[3]—most Scholars chose a single category corresponding to the one they selected on their GMS application form. A small percent in each racial minority group chose a minority race in addition to or other than the one they selected on the GMS application form. The most frequent race combinations were White with one or more of the minority races. Close to 7 percent of the African Americans, 15 percent of the American Indians/Alaska Natives, and 3 percent of the Asian/Pacific Islanders indicated that they were also White.

Of the scholarship program's four minority groups, Hispanic American Scholars are the most diverse in terms of the five race categories. About 94 percent of Scholars in this group are White, 10 percent American Indian/Alaska Native, 9 percent Black/African American, and 3 percent Asian, alone or in combination with other races.

[3] These options correspond to the five race categories of the U.S. Office of Management and Budget. Each option is displayed as a separate entry in Exhibit 2. Since the Scholars were allowed to select more than a single category, the sum of the percents across categories will be greater than 100.

Exhibit 2. Racial/Ethnic Background of the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
American Indian or Alaska Native	Yes	23 (0.2)	21 (0.4)	23 (0.2)	2 (0.3)	100 (0.0)	0 (0.0)	10 (0.9)
	No	77 (0.2)	79 (0.4)	77 (0.2)	98 (0.3)	0 (0.0)	100 (0.0)	90 (0.9)
Native Hawaiian	Yes	1 (0.1)	0 (0.0)	2 (0.2)	1 (0.2)	1 (0.3)	4 (0.5)	0 (0.0)
	No	99 (0.1)	100 (0.0)	98 (0.2)	99 (0.2)	99 (0.3)	96 (0.5)	100 (0.0)
Asian	Yes	20 (0.1)	21 (0.2)	20 (0.2)	1 (0.2)	0 (0.0)	97 (0.4)	3 (0.5)
	No	80 (0.1)	79 (0.2)	80 (0.2)	99 (0.2)	100 (0.0)	3 (0.4)	97 (0.5)
Black/African American	Yes	48 (0.2)	49 (0.5)	47 (0.2)	100 (0.0)	1 (0.6)	1 (0.2)	9 (0.8)
	No	52 (0.2)	51 (0.5)	53 (0.2)	0 (0.0)	99 (0.6)	99 (0.2)	91 (0.8)
White	Yes	20 (0.5)	21 (1.0)	19 (0.5)	7 (0.5)	15 (1.6)	3 (0.4)	94 (0.7)
	No	80 (0.5)	79 (1.0)	81 (0.5)	93 (0.5)	85 (1.6)	97 (0.4)	6 (0.7)
Hispanic ethnicity	Yes	36 (0.2)	39 (0.4)	34 (0.1)	1 (0.2)	4 (0.9)	0 (0.0)	100 (0.1)
	No	64 (0.2)	61 (0.4)	66 (0.1)	99 (0.2)	96 (0.9)	100 (0.0)	0 (0.1)
Hispanic country of origin	Mexican or Chicano	61 (0.8)	62 (1.2)	61 (1.0)	0 (0.0)	100 (0.0)	0 (0.0)	61 (0.8)
	Puerto Rican	8 (0.4)	10 (0.7)	7 (0.5)	0 (0.0)	0 (0.0)	0 (0.0)	8 (0.4)
	Cuban	4 (0.3)	5 (0.5)	3 (0.3)	0 (0.0)	0 (0.0)	0 (0.0)	4 (0.3)
	Other Hispanic	27 (0.7)	23 (1.0)	29 (0.9)	100 (0.0)	0 (0.0)	0 (0.0)	27 (0.7)
Asian origin	Bangladeshi	2 (0.4)	2 (0.7)	2 (0.4)	0 (0.0)	0 (0.0)	2 (0.4)	0 (0.0)
	Cambodian	3 (0.4)	2 (0.7)	3 (0.5)	0 (0.0)	0 (0.0)	3 (0.4)	0 (0.0)
	Chinese	30 (1.1)	40 (2.2)	26 (1.2)	0 (0.0)	0 (0.0)	31 (1.1)	0 (0.0)
	Filipino	4 (0.5)	0 (0.0)	6 (0.7)	50 (12.1)	0 (0.0)	3 (0.4)	34 (7.7)
	Hmong	7 (0.6)	2 (0.7)	9 (0.8)	0 (0.0)	0 (0.0)	7 (0.6)	0 (0.0)
	Indian	6 (0.6)	9 (1.3)	5 (0.6)	0 (0.0)	0 (0.0)	7 (0.6)	0 (0.0)
	Japanese	1 (0.3)	2 (0.7)	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.3)	0 (0.0)
	Korean	16 (0.9)	19 (1.8)	15 (1.0)	50 (12.1)	0 (0.0)	16 (0.9)	0 (0.0)
	Laotian	2 (0.4)	2 (0.7)	2 (0.4)	0 (0.0)	0 (0.0)	2 (0.4)	0 (0.0)
	Nepali	1 (0.3)	2 (0.7)	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.3)	0 (0.0)
	Pakistani	1 (0.2)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.2)	0 (0.0)
	Thai	1 (0.2)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.2)	0 (0.0)
	Vietnamese	21 (1.0)	14 (1.6)	24 (1.2)	0 (0.0)	0 (0.0)	22 (1.0)	0 (0.0)
	Other	4 (0.4)	5 (0.9)	3 (0.5)	0 (0.0)	0 (0.0)	2 (0.4)	66 (7.7)
Native Hawaiian origin	Hawaiian	44 (5.0)	0 (0.0)	44 (5.0)	0 (0.0)	0 (0.0)	67 (5.3)	0 (0.0)
	Guamanian or Chamorro	11 (3.7)	0 (0.0)	11 (3.7)	50 (12.5)	0 (0.0)	0 (0.0)	0 (0.0)
	Tongan	33 (4.9)	0 (0.0)	33 (4.9)	0 (0.0)	100 (0.0)	33 (5.3)	0 (0.0)
	Other	11 (3.7)	0 (0.0)	11 (3.7)	50 (12.5)	0 (0.0)	0 (0.0)	0 (0.0)

Exhibit 3. Tribal Affiliations of the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
American Indian tribe	Absentee-Shawnee	1 (0.5)	3 (1.8)	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)	0 (0.0)
	Aleut	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Assiniboine & Sioux Tribes of the Fort Peck Indian Reserve	2 (0.6)	3 (1.8)	1 (0.4)	0 (0.0)	2 (0.6)	0 (0.0)	0 (0.0)
	Athabascan	2 (0.6)	3 (1.8)	1 (0.4)	0 (0.0)	2 (0.6)	0 (0.0)	0 (0.0)
	Blackfeet	2 (0.6)	6 (2.0)	1 (0.3)	14 (4.4)	1 (0.6)	0 (0.0)	14 (3.8)
	Cherokee (Unspecified)	13 (1.3)	15 (3.5)	13 (1.2)	29 (5.9)	13 (1.4)	0 (0.0)	14 (3.8)
	Cherokee Nation, Oklahoma	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Cheyenne River Sioux	3 (0.5)	0 (0.0)	4 (0.7)	0 (0.0)	3 (0.6)	0 (0.0)	0 (0.0)
	Chickasaw	2 (0.6)	3 (1.8)	1 (0.3)	0 (0.0)	1 (0.6)	0 (0.0)	14 (3.8)
	Chippewa	1 (0.5)	3 (1.8)	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)	0 (0.0)
	Choctaw	7 (0.9)	3 (1.8)	8 (1.0)	14 (4.7)	6 (0.9)	0 (0.0)	0 (0.0)
	Comanche	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Crow Creek Sioux	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Duckwater Shoshone	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Inupiaq	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Klamath Res. (Oregon)	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Lumbee Tribe (North Carolina)	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Lummi	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Makah	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Menominee	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Mississippi Band of Choctaw	1 (0.4)	0 (0.0)	2 (0.5)	0 (0.0)	2 (0.4)	0 (0.0)	0 (0.0)
	Multiple Tribes	5 (0.8)	3 (1.8)	6 (0.8)	14 (4.7)	4 (0.8)	0 (0.0)	14 (3.8)
	Muscogee (Creek)	3 (0.8)	6 (2.5)	2 (0.5)	0 (0.0)	4 (0.9)	0 (0.0)	0 (0.0)
	Navajo	26 (1.7)	31 (4.6)	24 (1.6)	0 (0.0)	28 (1.9)	0 (0.0)	14 (3.4)
	Oglala Sioux	4 (0.7)	3 (1.8)	4 (0.7)	0 (0.0)	4 (0.8)	0 (0.0)	0 (0.0)
	Omaha	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Onondaga	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Osage	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Paiute	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Passamaquoddy	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Pueblo of Isleta	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Pueblo of Santo Domingo	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Salt River Pima	1 (0.5)	3 (1.8)	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)	0 (0.0)
	Sisseton-Wahpeton Sioux	1 (0.4)	0 (0.0)	2 (0.5)	0 (0.0)	2 (0.4)	0 (0.0)	0 (0.0)
	Spokane	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Tlingit	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Turtle Mountain Band of Chippewa	3 (0.8)	6 (2.5)	2 (0.5)	0 (0.0)	4 (0.9)	0 (0.0)	0 (0.0)
	Yurok	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Zuni	1 (0.4)	0 (0.0)	2 (0.5)	0 (0.0)	2 (0.4)	0 (0.0)	0 (0.0)
	Apache (Unspecified)	1 (0.2)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)	0 (0.0)	14 (3.8)
	Aztec	1 (0.2)	2 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	14 (3.4)
	Gros Ventre	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Leech Lake Band of Ojibwe	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Lakota Sioux	2 (0.6)	3 (1.8)	1 (0.4)	0 (0.0)	2 (0.6)	0 (0.0)	0 (0.0)
	Coos (unspecified)	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Flathead	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Nomalacki & Wylacki	1 (0.3)	0 (0.0)	1 (0.4)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Pequouet	1 (0.2)	2 (0.8)	0 (0.0)	14 (4.4)	0 (0.0)	0 (0.0)	0 (0.0)
	Haitian Native American	1 (0.2)	0 (0.0)	1 (0.3)	14 (4.7)	0 (0.0)	0 (0.0)	0 (0.0)

Like the Hispanic American Scholars and non-recipients in earlier cohorts, Cohort V Hispanic American Scholars were far less likely to answer the racial background question than their minority counterparts. Close to 70 percent of the survey respondents in this subpopulation chose not to answer the question compared to less than 2 percent of respondents in the other minority groups (not shown in the exhibit). It appears that these Scholars considered themselves to be some race other than the ones listed in the racial background question and without any other options available simply chose not to answer the question.

Due to the high level of non-response, the estimates for the Hispanic American subpopulation on this question should be interpreted with care despite the relatively small size of their standard errors.[4]

Hispanic ethnicity. Only a small share of the Scholars in the three racial minority groups identified themselves as Hispanic or Latino. This group includes about 1 percent of the African Americans and 4 percent of the American Indians/Alaska Natives.

Country of origin. Scholars who indicated that they were Asian, Native Hawaiian or Other Pacific Islander, or Hispanic or Latino were asked to identify the country or island of their origin in a drop-down list or to key in their answer if it was not on the list. As a whole, the Scholars reported origins in over 30 different countries and islands from around the world.

We find in Exhibit 2 that nearly three-fourths of the Hispanic American Scholars have origins in Mexico, Puerto Rico, and Cuba. Sixty-one percent of Hispanic American Scholars are of Mexican or Chicano descent. Another 8 percent are of Puerto Rican descent, and finally, 4 percent are Cuban. The remaining Scholars are of other Hispanic origins.

Vietnam, China, and Korea account for the ethnic origins of about 69 percent of the Asian Scholars.[5] Twenty-two percent have origins in Vietnam, another 31 percent are Chinese, and 16 percent are Korean. The remaining Scholars are of other Asian origins.

Finally, we see that about two-thirds of the Pacific Islanders in the Asian/Pacific Islander group of Scholars are Hawaiian, and one-third of these Scholars are Tongan.[6]

Tribe. Scholars who indicated that they were American Indians or Alaska Natives were asked to key in the name of their tribe. NORC coded their responses into categories according to a framework provided by the American Indian Graduate Center, a partner organization in the GMS program.

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[4] Also, please note that the high level of non-response among Hispanic Americans on the race question will affect the estimates of the percents of American Indians/Alaska Natives, Native Hawaiians, Asians, Black/African Americans, and Whites in the population of Scholars as a whole. Due to the high level of non-response, the size of the Scholar population in those calculations is substantially reduced. In this case, the survey weights will not be effective in reducing the bias associated with non-response among the Hispanic Americans.

[5] Asian Scholars include all Scholars in the Asian/Pacific Islander subpopulation of Scholars who identified themselves as Asian on the survey.

[6] Pacific Islander Scholars include all Scholars in the Asian/Pacific Islander subpopulation of Scholars who identified themselves as Pacific Islanders in the survey.

As we see in Exhibit 3, Scholars in the American Indian/Alaska Native subpopulation are affiliated with over 40 different state and federally recognized American Indian and Alaska Native tribes. About 26 percent belong to the Navajo tribe, 14 percent to the Cherokee (Cherokee Nation of Oklahoma or unspecified), 7 percent to the Choctaw, and 5 percent to multiple tribes. The remaining Scholars are distributed in small numbers across the other tribes shown in Exhibit 3.

2.1.2 Geographic Background

A question in the baseline survey asked the Scholars where they graduated from high school. According to their responses to this question, we classified the Scholars into one of the Census Bureau's nine divisions of the states or as "outside the United States." (See Appendix H for a listing of the states in each division.)

We see in Exhibit 4 that almost none of the Scholars graduated from high school outside of the 50 states in one of the U.S. possessions or in a foreign country. The remaining Scholars completed high school in the United States.

Exhibit 4. Geographic Background of the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Census Division of HS	New England	1 (0.1)	2 (0.2)	1 (0.1)	2 (0.3)	0 (0.0)	1 (0.3)	2 (0.2)
	Middle Atlantic	9 (0.3)	11 (0.5)	8 (0.3)	9 (0.6)	1 (0.3)	18 (0.9)	8 (0.4)
	Midwest	5 (0.3)	5 (0.5)	6 (0.3)	10 (0.6)	3 (0.7)	4 (0.4)	2 (0.2)
	West North Central	5 (0.3)	4 (0.5)	5 (0.3)	4 (0.4)	13 (1.3)	8 (0.6)	1 (0.2)
	South Atlantic	19 (0.4)	19 (0.7)	18 (0.5)	30 (0.9)	2 (0.5)	15 (0.9)	16 (0.6)
	East South Central	5 (0.3)	4 (0.4)	6 (0.3)	14 (0.7)	1 (0.3)	2 (0.3)	0 (0.0)
	West South Central	21 (0.5)	21 (1.0)	21 (0.5)	20 (0.8)	29 (1.9)	8 (0.7)	24 (0.7)
	Mountain	12 (0.4)	13 (0.8)	11 (0.4)	0 (0.1)	40 (2.1)	4 (0.4)	14 (0.5)
	Pacific	23 (0.4)	22 (0.8)	23 (0.5)	11 (0.6)	11 (1.2)	40 (1.2)	33 (0.7)
	Outside the U.S.	0 (0.1)	1 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.1)

Close to two-thirds of the Scholars graduated from secondary school in the Pacific, West South Central, or South Atlantic divisions of the country. Another 12 percent graduated from one of the Mountain states. Less than 10 percent of the Scholars completed high school in each of the five remaining divisions.

The distribution of Scholars across the divisions is similar for males and females. However, there are slightly larger shares of males than females from the New England, Middle Atlantic, and Mountain divisions and slightly larger shares of females than males from the East South Central division.

2.1.3 Socio-Economic Background

The baseline survey asked the Scholars to supply information on the educational attainment levels of their parents. Exhibit 5 displays their answers to this set of questions as estimates of the population and subpopulation percents.

Educational attainment of the parents. As Exhibit 5 shows, over three-fourths of the Scholars' parents earned at least a high school diploma or its equivalent (e.g., GED). Twenty-one percent of their fathers and 27 percent of their mothers completed at least a bachelor's degree, while 9 and 8 percent, respectively, completed advanced degrees.[7] The percent of Scholars' mothers earning at least a bachelor's degree is significantly higher than for the Scholars' fathers; conversely, the percent of Scholars' mothers earning advanced degrees is significantly lower than for the fathers. The shares of fathers and mothers earning high school degrees do not differ significantly.

Exhibit 5 shows that the educational attainment levels of the Scholars' parents vary by minority group. Overall, the parents of the Hispanic American Scholars tend to show lower levels of educational attainment compared to the parents of the other minority groups. The fathers of the American Indian/Alaska Native Scholars were the most likely to have attained at least a high school degree (92 percent), followed by the fathers of the African American Scholars (88 percent). For the Asian/Pacific Islanders, while the level at which the fathers earned at least high school degrees (80 percent) was below that of both the American Indians/Alaska Natives and the African/Americans, it was substantially above that of the Hispanic American fathers (56 percent). The pattern for the Scholars' mothers is somewhat similar. At the high school level, the mothers of the American Indian/Alaska Native Scholars were the most likely to have earned at least a high school degree (97 percent), followed by the mothers of the African American Scholars (90 percent). The mothers of the Asian/Pacific Islander and Hispanic American Scholars were the least likely to have completed at least a high school degree (61 percent for both groups).

[7] To obtain the percent completing at least a high school degree, we took the compliment of the percent with less than a high school degree (100 minus the percent with less than a high school degree). We computed the percent completing at least a bachelor's degree by summing the percents earning a bachelor's, master's, or doctoral degree as their highest degree. Similarly, we obtained the percent with advanced degrees by summing the percents with a master's or doctoral degree as their highest degree. Note that the summed percents may not agree with those based on the percents in the exhibits due to rounding error.

Exhibit 5. Educational Attainment of the Cohort V Scholars' Parents for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Father's highest level of educational attainment	Less than high school	24 (0.4)	24 (0.8)	24 (0.5)	12 (0.7)	8 (1.2)	20 (1.0)	44 (0.8)
	GED	4 (0.2)	4 (0.4)	4 (0.3)	4 (0.4)	7 (0.8)	2 (0.4)	3 (0.3)
	High school graduation	26 (0.5)	26 (1.0)	26 (0.6)	28 (0.9)	32 (2.0)	30 (1.1)	19 (0.6)
	Some college	25 (0.5)	22 (1.0)	26 (0.6)	30 (0.9)	37 (2.1)	18 (1.0)	19 (0.6)
	Bachelor's degree	12 (0.4)	13 (0.8)	11 (0.4)	14 (0.7)	11 (1.5)	10 (0.8)	10 (0.5)
	Master's degree or equivalent	7 (0.3)	7 (0.6)	7 (0.3)	9 (0.6)	4 (1.0)	13 (0.9)	5 (0.3)
	Doctorate	2 (0.2)	2 (0.4)	2 (0.2)	3 (0.3)	1 (0.6)	6 (0.6)	0 (0.1)
Mother's highest level of educational attainment	Less than high school	23 (0.4)	22 (0.7)	24 (0.5)	10 (0.6)	3 (0.6)	39 (1.2)	39 (0.8)
	GED	4 (0.2)	4 (0.4)	4 (0.3)	2 (0.3)	7 (1.0)	1 (0.2)	7 (0.4)
	High school graduation	20 (0.5)	23 (1.0)	19 (0.5)	22 (0.8)	25 (1.8)	25 (1.1)	14 (0.5)
	Some college	26 (0.5)	20 (0.9)	28 (0.6)	32 (0.9)	34 (1.8)	13 (0.8)	21 (0.6)
	Bachelor's degree	19 (0.5)	23 (1.0)	17 (0.5)	25 (0.9)	22 (1.8)	12 (0.8)	13 (0.5)
	Master's degree or equivalent	7 (0.3)	7 (0.6)	8 (0.4)	9 (0.6)	9 (1.2)	8 (0.7)	5 (0.3)
	Doctorate	1 (0.1)	1 (0.2)	0 (0.0)	1 (0.2)	0 (0.0)	2 (0.3)	0 (0.1)
Father's attainment (HS degree)	At least a HS degree	76 (0.4)	76 (0.8)	76 (0.5)	88 (0.7)	92 (1.2)	80 (1.0)	56 (0.8)
	Less than a HS degree	24 (0.4)	24 (0.8)	24 (0.5)	12 (0.7)	8 (1.2)	20 (1.0)	44 (0.8)
Father's attainment (bachelor's degree)	At least a bachelor's degree	21 (0.5)	23 (1.0)	20 (0.5)	25 (0.9)	16 (1.7)	29 (1.1)	15 (0.6)
	Less than a bachelor's degree	79 (0.5)	77 (1.0)	80 (0.5)	75 (0.9)	84 (1.7)	71 (1.1)	85 (0.6)
Father's attainment (advanced degree)	Advanced degree	9 (0.3)	10 (0.7)	9 (0.4)	11 (0.7)	5 (1.1)	19 (1.0)	5 (0.3)
	Less than an advanced degree	91 (0.3)	90 (0.7)	91 (0.4)	89 (0.7)	95 (1.1)	81 (1.0)	95 (0.3)
Mother's attainment (HS degree)	At least a HS degree	77 (0.4)	78 (0.7)	76 (0.5)	90 (0.6)	97 (0.6)	61 (1.2)	61 (0.8)
	Less than a HS degree	23 (0.4)	22 (0.7)	24 (0.5)	10 (0.6)	3 (0.6)	39 (1.2)	39 (0.8)
Mother's attainment (bachelor's degree)	At least a bachelor's degree	27 (0.5)	31 (1.1)	25 (0.6)	35 (1.0)	30 (2.0)	22 (1.0)	19 (0.6)
	Less than a bachelor's degree	73 (0.5)	69 (1.1)	75 (0.6)	65 (1.0)	70 (2.0)	78 (1.0)	81 (0.6)
Mother's attainment (advanced degree)	Advanced degree	8 (0.3)	8 (0.6)	8 (0.4)	10 (0.6)	9 (1.2)	9 (0.7)	6 (0.4)
	Less than an advanced degree	92 (0.3)	92 (0.6)	92 (0.4)	90 (0.6)	91 (1.2)	91 (0.7)	94 (0.4)

The fathers of the American Indian/Alaska Native (16 percent) and the Hispanic American (15 percent) Scholars were the least likely to have earned at least a bachelor's degree. In comparison, the fathers of the Asian/Pacific Islanders were most likely to have earned at least a bachelor's degree (29 percent), followed by the fathers of the African American Scholars (25 percent).

With regard to the rates at which mothers were attaining at least a bachelor's degree, the mothers of the Hispanic Americans and Asian/Pacific Islanders show the lowest rate (19 percent and 22 percent, respectively). They were both less likely than the mothers of the African American (35 percent) and the American Indian/Alaska Native (30 percent) Scholars to have attained at least a bachelor's degree.

The parents of the Hispanic American Scholars have the lowest levels of attainment of all four minority groups. Sixty-one percent or less of the Hispanic American Scholars' parents completed at least a high school degree, 19 percent or less earned at least a bachelor's degree, and 6 percent or less, a graduate or professional degree.

2.1.4 Educational Background and Secondary School Experiences

As part of the baseline survey, the Scholars were asked several questions regarding their educational background and secondary school experiences. These questions included items on the characteristics of the high school they attended, number of advanced placement (AP) exams taken, and years of math and science they took in high school.

Race majority of students in high school classes. We find in Exhibit 6 that over 40 percent of the Scholars agreed or strongly agreed that the classes they took in high school were with students mostly of the same race they were; 25 percent strongly agreed. Fifty-seven percent of Scholars either disagreed or strongly disagreed that they were in the race majority in their high school classes. This suggests that most Scholars were in classes with sizable shares of students from racial/ethnic backgrounds other than their own.

Exhibit 6 also suggests that the relative racial composition of the Scholars' high school classes varied by minority group. According to their self-reports, American Indian/Alaska Native Scholars were more likely than other minority groups to strongly agree (43 percent) that they were in classes where they were in the race majority. In comparison, 26 percent of African Americans and 25 percent of Hispanic American Scholars strongly agreed. At 5 percent, Asian/Pacific Islanders were the least likely of all the minority groups to report that they took high school classes with students mostly of their own race.

Advanced placement exams. As shown in Exhibit 6, about three-fourths of the Scholars took at least one advanced placement exam while in high school. About 64 percent took two or more advanced placement exams, 49 percent, three or more, and 35 percent, four or more sometime during their high school careers.[8]

[8] To find the percent who took at least one advanced placement exam, we took the compliment of the percent who took none (100 minus the percent who took none). The standard errors of both estimates are the same. To find the percent who took two or more exams, we summed the percents who took two, three, or four or more exams. Similarly, to find the percent who took three or more exams, we added the percents who took three or four or more exams.

The minority groups differ markedly in the number of advanced placement exams taken. Asian/Pacific Islanders and Hispanic Americans were more likely than the other minority groups to take at least one advanced placement exam; 86 percent of Asian/Pacific Islanders and 85 percent of Hispanic Americans took at least one exam compared to 71 and 49 percent, respectively, of African American and American Indian/Alaska Native Scholars. With just over half not taking any advanced placement exams at all, American Indians/Alaska Natives were the least likely of any minority group to take at least one advanced placement exam.

Exhibit 6. Educational Background and Secondary School Experiences of the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Race majority of the students in nominee's HS classes	Strongly Disagree	32 (0.5)	35 (1.0)	31 (0.6)	37 (1.0)	15 (1.6)	50 (1.2)	28 (0.7)
	Disagree	25 (0.5)	28 (1.0)	24 (0.6)	23 (0.8)	20 (1.7)	32 (1.1)	27 (0.7)
	Agree	17 (0.4)	14 (0.7)	19 (0.5)	14 (0.7)	22 (1.5)	12 (0.8)	20 (0.6)
	Strongly Agree	25 (0.5)	23 (1.0)	26 (0.6)	26 (0.9)	43 (2.1)	5 (0.5)	25 (0.7)
No. of AP exams taken in high school	None	25 (0.5)	24 (1.0)	25 (0.6)	29 (0.9)	51 (2.2)	14 (0.8)	15 (0.6)
	One	12 (0.4)	10 (0.7)	12 (0.5)	16 (0.7)	18 (1.6)	5 (0.5)	8 (0.4)
	Two	15 (0.4)	14 (0.8)	15 (0.5)	15 (0.7)	16 (1.5)	12 (0.8)	15 (0.6)
	Three	14 (0.4)	14 (0.7)	14 (0.4)	15 (0.7)	6 (1.1)	16 (0.9)	14 (0.5)
	Four or more	35 (0.5)	38 (0.9)	34 (0.6)	25 (0.9)	8 (1.3)	54 (1.2)	48 (0.8)
Attended a religious high school	Yes	7 (0.3)	8 (0.6)	7 (0.3)	7 (0.5)	9 (1.2)	2 (0.4)	8 (0.4)
	No	93 (0.3)	92 (0.6)	93 (0.3)	93 (0.5)	91 (1.2)	98 (0.4)	92 (0.4)
Attended a private high school	Yes	10 (0.4)	11 (0.7)	10 (0.4)	9 (0.6)	17 (1.5)	5 (0.5)	10 (0.5)
	No	90 (0.4)	89 (0.7)	90 (0.4)	91 (0.6)	83 (1.5)	95 (0.5)	90 (0.5)
No. of years of math completed in HS	None	0 (0.0)	0 (0.1)	0 (0.0)	0 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)
	One	0 (0.0)	0 (0.1)	0 (0.0)	0 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)
	Two	1 (0.1)	1 (0.1)	1 (0.2)	1 (0.2)	2 (0.5)	1 (0.3)	1 (0.1)
	Three	12 (0.4)	10 (0.6)	14 (0.5)	12 (0.6)	15 (1.4)	9 (0.7)	13 (0.5)
	Four or more	86 (0.4)	89 (0.7)	85 (0.5)	87 (0.7)	82 (1.4)	90 (0.7)	86 (0.5)
No. of years of science completed in HS	One	0 (0.0)	0 (0.0)	0 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.1)
	Two	3 (0.2)	3 (0.3)	3 (0.2)	1 (0.2)	5 (0.7)	1 (0.2)	5 (0.3)
	Three	24 (0.5)	24 (1.0)	24 (0.6)	23 (0.8)	24 (1.8)	20 (0.9)	26 (0.7)
	Four or more	73 (0.5)	73 (1.0)	73 (0.6)	75 (0.8)	72 (1.9)	79 (1.0)	68 (0.7)

Asian/Pacific Islanders were more likely than their minority counterparts to complete four or more advanced placement exams, with 54 percent of the Asian/Pacific Islanders in that category. At 48 percent, Hispanic Americans were more likely than both African Americans (25 percent) and American Indians/Alaska Natives (8 percent) to take four or more advanced placement exams. American

Indians/Alaska Natives were the least likely of any minority group to take four or more advanced placement exams.

Attendance at religious or private high schools. The large majority of Scholars attended public high schools; only 10 percent were enrolled in private schools, and 7 percent in religious schools (see Exhibit 6).

Overall, Asian/Pacific Islander Scholars were less likely than were their minority counterparts to attend religious high schools. Only 2 percent of Asian/Pacific Islanders attended a religious high school compared to 7 percent of African Americans, 9 percent of American Indians/Alaska Natives, and 8 percent of Hispanic Americans.

About 17 percent of American Indian/Alaska Native Scholars attended a private high school, significantly more than the other three minority groups. About 9 percent of African Americans and 10 percent of Hispanic Americans attended private high schools. At 5 percent, Asian/Pacific Islanders were the least likely of the minority groups to attend a private high school.

Years of math and science in high school. As Exhibit 6 shows, almost all (98 percent) of the Scholars completed at least three years of math while in high school. Many Scholars (86 percent) took four or more years of math.[9]

In terms of the number of years taken, the self-reported high school science backgrounds of the Scholars are nearly as strong as their math backgrounds. Ninety-seven percent reported taking three or more years of science in high school, and 73 percent said they took four or more years of science.

As a whole, females were less likely than their male counterparts to take four or more years of math, but there was no significant difference by gender in the proportions completing four or more years of science while in high school.

With regard to the four minority groups, Asian/Pacific Islander Scholars were more likely than the other minorities to take four or more years of math, followed by African Americans and Hispanic Americans, who did not differ significantly from one another. American Indians/Alaska Natives were less likely than the other minority groups to take four or more years of math.

Asian/Pacific Islanders were more likely than the other minority groups to take four or more years of science in high school. African Americans were similar to American Indians/Alaska Natives in the percentages that took four or more years of science. Although African Americans were more likely than Hispanic Americans to take four or more years of science, American Indians/Alaska Natives and Hispanic Americans did not differ significantly.

[9] To find the percent of Scholars who took three or more years of math or science, we summed the percents who took three or four or more years of math or science.

2.1.5 Educational Aspirations and Expectations

As part of the baseline survey, the Scholars were asked to provide information on their educational aspirations and expectations. Their answers to this question are shown in Exhibit 7.

At the time of the baseline survey, nearly all of the Scholars (rounded to 100 percent in Exhibit 7) expected to complete at least a four-year bachelor's degree, and most (92 percent) expected to go on to complete an advanced degree.[10]

Exhibit 7. Educational Aspirations of the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Educational aspirations (highest level of aspiration)	Less than two years of college	0 (0.0)	0 (0.0)	0 (0.1)	0 (0.0)	1 (0.3)	0 (0.0)	0 (0.0)
	Two or more years of college	0 (0.1)	0 (0.0)	0 (0.1)	0 (0.0)	2 (0.4)	0 (0.0)	0 (0.0)
	Bachelor's degree	7 (0.3)	6 (0.7)	7 (0.4)	5 (0.4)	19 (1.7)	5 (0.5)	5 (0.4)
	Post-baccalaureate certificate	1 (0.1)	0 (0.0)	1 (0.1)	0 (0.1)	2 (0.4)	0 (0.0)	1 (0.1)
	Master's degree (M.A., M.S., M.B.A., etc.)	30 (0.5)	30 (1.1)	30 (0.6)	29 (0.9)	29 (2.0)	29 (1.1)	32 (0.7)
	First professional degree (M.D., J.D., D.D.S., O.D.)	24 (0.5)	26 (1.0)	23 (0.6)	24 (0.9)	19 (1.7)	31 (1.2)	23 (0.7)
	Doctoral degree (PH.D., ED.D., D.P.H., etc.)	38 (0.6)	38 (1.1)	39 (0.7)	42 (1.0)	29 (2.0)	35 (1.2)	40 (0.8)
Educational aspirations (bachelor's degree)	At least a bachelor's degree	100 (0.1)	100 (0.0)	99 (0.1)	100 (0.0)	97 (0.5)	100 (0.0)	100 (0.0)
	Less than a bachelor's degree	0 (0.1)	0 (0.0)	1 (0.1)	0 (0.0)	3 (0.5)	0 (0.0)	0 (0.0)
Educational aspirations (master's degree)	At least a master's degree	92 (0.3)	94 (0.7)	91 (0.4)	95 (0.4)	77 (1.8)	95 (0.5)	94 (0.4)
	Less than a master's degree	8 (0.3)	6 (0.7)	9 (0.4)	5 (0.4)	23 (1.8)	5 (0.5)	6 (0.4)
Educational aspirations (first professional or doctoral degree)	First professional or doctoral degree	62 (0.6)	64 (1.1)	61 (0.7)	66 (1.0)	48 (2.2)	66 (1.2)	62 (0.8)
	Less than a professional or doctoral degree	38 (0.6)	36 (1.1)	39 (0.7)	34 (1.0)	52 (2.2)	34 (1.2)	38 (0.8)

[10] To find the percent expecting to complete at least a bachelor's degree, we summed the percents expecting to complete a post-baccalaureate certificate or a bachelor's, master's, first professional, or doctoral degree as their highest level of educational attainment. The percent aspiring to obtain an advanced degree was computed by summing the percents expecting to complete a master's, first professional, or doctoral degree as their highest degree. Note that the summed percents may not agree with those based on the percents in the exhibits due to rounding error.

African Americans, Asian/Pacific Islanders, and Hispanic Americans all had high expectations of completing an advanced degree. Ninety-five percent of both the African Americans and the Asian/Pacific Islanders, along with 94 percent of Hispanic Americans aspired to complete at least a master's degree. At 77 percent, the American Indians/Alaska Natives were the least likely to believe they would earn at least a master's degree. In terms of first professional or doctoral degrees, African Americans and Asian/Pacific Islanders were the most likely to believe that they would complete a professional or doctoral degree (66 percent for both groups), followed closely by Hispanic Americans (62 percent). American Indians/Alaska Natives, at 48 percent, were the least likely of any minority group to believe that they would go on to earn a first professional or doctoral degree.

2.2 Who Are the Cohort V Non-Recipients? How Are They Similar to or Different from the Cohort V Scholars?

The Cohort V non-recipients who were asked to participate in the study are a representative sample of GMS nominees who, like the Scholars, applied to the program for aid and met all the basic requirements of the screening phase, but did not become Scholars. Some were disqualified or eliminated during the Scholar verification/confirmation stage. Others were not asked to go on to that stage. Still others were offered a scholarship but declined to accept it. Altogether, the non-recipient population consists of 3,464 nominees and includes more than twice as many females as males.[11]

We see in Exhibit 8 that this population of non-recipients, like that of the Scholars, is made up of four minority groups: African Americans, American Indians/Alaska Natives (referred to as American Indians in the exhibits), Asian/Pacific Islanders, and Hispanic Americans. The number of non-recipients varies by minority group, with more than 1,000 African Americans and fewer than 400 American Indians/Alaska Natives.

Exhibit 8. Number of Cohort V Non-Recipients by Minority Group and Gender as Reported on the GMS Application Form

Minority Group	Gender		
	Males	Females	Total
African American	253	840(1)*	1093(1)*
American Indian	103	251	354
Asian/Pacific Islander	191	494	685
Hispanic American	430	901	1331
Total Population	977	2486	3463(1)*

Source: GMS application form

*The number in parentheses indicates that one population member was known to be deceased at the time of the baseline survey and has been removed from the counts in the exhibit.

From this population of non-recipients, NORC selected a stratified random sample of 1,333 nominees and asked them to participate in the longitudinal survey. (Refer to Appendix A for a description of the sample

[11] The population total includes one person who was known to be deceased at the time of the baseline survey.

design for the non-recipients.) Below we discuss their responses to the same set of questions asked of the Scholars and compare their answers with those of the Scholars.

In the exhibits that follow, the non-recipients are classified into minority groups and by gender with regard to the information they supplied on the GMS application form.

2.2.1 Racial and Ethnic Background

Like the Scholars, the non-recipients were asked to supply detailed information on their racial backgrounds and their ethnic origins. Their responses to this set of questions are presented in Exhibits 9 and 10 as population and subpopulation percents along with their corresponding standard errors.

Race. When asked to classify themselves into one or more of the five race categories, most of the non-recipients in the three *racial* minority groups—African Americans, American Indians/Alaska Natives, and Asian/Pacific Islanders—selected a single category corresponding to the one they selected on the GMS application form. A small proportion of non-recipients in each of the groups reported that they were a minority race other than or in addition to the one they specified on their GMS application form, and a small proportion of the African Americans and Asian/Pacific Islanders reported that they were also White. On the other hand, a relatively large share of the American Indians/Alaska Natives (21 percent) indicated that they were also White.

Overall, the racial composition of the four minority groups is in close accord with that of the Scholar population.

Like the Scholars, the Hispanic American non-recipients were less likely to answer the racial background question than members of the other three minority groups. About 60 percent of the Hispanic American respondents declined to answer compared to less than 3 percent of respondents in the other groups (not shown in the exhibit). Again, the estimates for the Hispanic American subpopulation on the racial background question should be interpreted with care due to the high level of non-response on that question.[12]

Hispanic ethnicity. As was the case for the Scholars, only a small percent of the non-recipients in the three *racial* groups identified themselves as Hispanic or Latino.

Country of origin. Like the Scholars, the non-recipients reported origins in more than 30 different countries and islands around the world. About 70 percent of the Asian non-recipients have ethnic origins in Vietnam,

[12] Also, please note that the high level of non-response among Hispanic Americans on the race question will affect the estimates of the percents of American Indians/Alaska Natives, Native Hawaiians, Asians, Black/African Americans, and Whites in the population of non-recipients as a whole. Due to the high level of non-response, the size of the non-recipient population in those calculations is substantially reduced. In this case, the survey weights will not be effective in reducing the bias associated with non-response among the Hispanic Americans.

China, Korea, or the Philippines.[13] About 27 percent are Chinese, 22 percent Vietnamese, 13 percent Korean, and another 8 percent are Filipino.

Three countries, Mexico, Puerto Rico, and Cuba, account for the ethnic origins of 73 percent of the Hispanic American subpopulation of non-recipients. About 60 percent of these non-recipients are of Mexican or

Exhibit 9. Racial/Ethnic Background of the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
American Indian or Alaska Native	Yes	16 (0.6)	16 (1.2)	16 (0.7)	3 (0.9)	98 (1.1)	1 (0.5)	6 (1.6)
	No	84 (0.6)	84 (1.2)	84 (0.7)	97 (0.9)	2 (1.1)	99 (0.5)	94 (1.6)
Native Hawaiian	Yes	1 (0.3)	1 (0.7)	1 (0.4)	0 (0.0)	0 (0.0)	5 (1.2)	1 (0.6)
	No	99 (0.3)	99 (0.7)	99 (0.4)	100 (0.0)	100 (0.0)	95 (1.2)	99 (0.6)
Asian	Yes	25 (0.5)	27 (1.2)	25 (0.5)	0 (0.0)	0 (0.0)	96 (1.1)	4 (1.4)
	No	75 (0.5)	73 (1.2)	75 (0.5)	100 (0.0)	100 (0.0)	4 (1.1)	96 (1.4)
Black/African American	Yes	42 (0.6)	35 (1.1)	45 (0.7)	100 (0.0)	2 (1.1)	1 (0.7)	3 (1.3)
	No	58 (0.6)	65 (1.1)	55 (0.7)	0 (0.0)	98 (1.1)	99 (0.7)	97 (1.3)
White	Yes	24 (1.1)	28 (2.3)	23 (1.3)	4 (1.0)	21 (3.6)	6 (1.3)	91 (2.1)
	No	76 (1.1)	72 (2.3)	77 (1.3)	96 (1.0)	79 (3.6)	94 (1.3)	9 (2.1)
Hispanic ethnicity	Yes	40 (0.3)	44 (0.4)	38 (0.4)	2 (0.7)	7 (2.2)	0 (0.0)	100 (0.0)
	No	60 (0.3)	56 (0.4)	62 (0.4)	98 (0.7)	93 (2.2)	100 (0.0)	0 (0.0)
Hispanic country of origin	Mexican or Chicano	60 (2.2)	62 (4.1)	59 (2.5)	17 (13.0)	74 (13.3)	0 (0.0)	60 (2.2)
	Puerto Rican	8 (1.2)	8 (2.3)	9 (1.4)	17 (13.0)	13 (10.2)	0 (0.0)	8 (1.2)
	Cuban	5 (0.9)	5 (1.8)	4 (1.0)	0 (0.0)	0 (0.0)	0 (0.0)	5 (0.9)
	Other Hispanic	27 (2.0)	25 (3.7)	28 (2.3)	67 (16.5)	13 (10.2)	0 (0.0)	27 (2.0)
Asian origin	Bangladeshi	1 (0.6)	2 (1.5)	1 (0.6)	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
	Cambodian	1 (0.5)	0 (0.0)	1 (0.7)	0 (0.0)	0 (0.0)	1 (0.5)	0 (0.0)
	Chinese	27 (2.6)	18 (4.5)	31 (3.1)	0 (0.0)	0 (0.0)	27 (2.6)	21 (16.4)
	Filipino	8 (1.5)	4 (2.0)	10 (2.0)	0 (0.0)	0 (0.0)	8 (1.6)	0 (0.0)
	Hmong	7 (1.4)	2 (1.5)	9 (1.9)	0 (0.0)	0 (0.0)	7 (1.5)	0 (0.0)
	Indian	7 (1.5)	9 (3.1)	6 (1.6)	0 (0.0)	0 (0.0)	7 (1.5)	0 (0.0)
	Japanese	4 (1.1)	2 (1.5)	5 (1.4)	0 (0.0)	0 (0.0)	4 (1.1)	0 (0.0)
	Korean	14 (2.0)	11 (3.7)	15 (2.4)	0 (0.0)	0 (0.0)	13 (2.0)	39 (18.9)
	Laotian	1 (0.7)	2 (1.5)	1 (0.8)	0 (0.0)	0 (0.0)	2 (0.7)	0 (0.0)
	Pakistani	3 (0.9)	5 (2.6)	1 (0.8)	0 (0.0)	0 (0.0)	3 (1.0)	0 (0.0)
	Thai	1 (0.7)	0 (0.0)	2 (0.9)	0 (0.0)	0 (0.0)	1 (0.7)	0 (0.0)
	Vietnamese	22 (2.4)	38 (5.5)	16 (2.5)	0 (0.0)	0 (0.0)	22 (2.4)	39 (18.9)
	Other	3 (1.1)	7 (2.9)	2 (0.9)	0 (0.0)	0 (0.0)	4 (1.1)	0 (0.0)
Native Hawaiian origin	Hawaiian	28 (11.5)	33 (24.1)	25 (12.7)	0 (0.0)	0 (0.0)	31 (12.4)	0 (0.0)
	Guamanian or Chamorro	19 (9.9)	33 (22.4)	13 (9.7)	0 (0.0)	0 (0.0)	21 (10.8)	0 (0.0)
	Fijian	19 (10.1)	33 (24.1)	13 (9.7)	0 (0.0)	0 (0.0)	21 (11.1)	0 (0.0)
	Maori	8 (5.9)	0 (0.0)	11 (8.2)	0 (0.0)	0 (0.0)	8 (6.5)	0 (0.0)
	Palauan	9 (7.0)	0 (0.0)	13 (9.7)	0 (0.0)	0 (0.0)	10 (7.7)	0 (0.0)
	Tongan	9 (7.0)	0 (0.0)	13 (9.7)	0 (0.0)	0 (0.0)	10 (7.7)	0 (0.0)
	Other	10 (7.9)	0 (0.0)	14 (10.9)	0 (0.0)	0 (0.0)	0 (0.0)	100 (0.0)

[13] Asian non-recipients include all non-recipients in the Asian/Pacific Islander subpopulation of non-recipients who identified themselves as Asian in the survey.

Exhibit 10. Tribal Affiliations of the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
American Indian tribe	Aleut	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Blackfeet	3 (1.3)	0 (0.0)	4 (1.8)	23 (12.3)	1 (0.8)	0 (0.0)	0 (0.0)
	Cherokee (Unspecified)	19 (3.2)	22 (5.8)	18 (3.8)	0 (0.0)	21 (3.5)	100 (0.0)	0 (0.0)
	Cheyenne-Arapaho	1 (0.8)	3 (2.7)	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	0 (0.0)
	Chickasaw	1 (0.7)	3 (2.2)	0 (0.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Chippewa	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Choctaw	10 (2.4)	11 (4.2)	9 (2.9)	0 (0.0)	11 (2.8)	0 (0.0)	0 (0.0)
	Confederated Salish & Kootenai	1 (0.7)	3 (2.2)	0 (0.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Echota Cherokees (Alabama)	1 (0.7)	3 (2.2)	0 (0.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Jicarilla	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Kickapoo Tribe of Kansas	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Klamath	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Lumbee Tribe (North Carolina)	2 (1.2)	0 (0.0)	3 (1.7)	0 (0.0)	2 (1.4)	0 (0.0)	0 (0.0)
	Mississippi Band of Choctaw	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Multiple Tribes	6 (2.1)	10 (4.7)	5 (2.1)	26 (13.8)	4 (1.7)	0 (0.0)	20 (15.0)
	Muscogee (Creek)	3 (1.2)	3 (2.2)	2 (1.5)	11 (9.3)	2 (1.1)	0 (0.0)	0 (0.0)
	Navajo	18 (3.1)	15 (5.2)	19 (3.8)	0 (0.0)	20 (3.5)	0 (0.0)	20 (15.0)
	Oglala Sioux	2 (1.1)	3 (2.2)	1 (1.2)	0 (0.0)	2 (1.3)	0 (0.0)	0 (0.0)
	Omaha	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Osage	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Pawnee	1 (0.7)	0 (0.0)	1 (0.9)	0 (0.0)	0 (0.0)	0 (0.0)	18 (13.8)
	Pueblo of Acoma	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Pueblo of San Felipe	1 (0.8)	3 (2.7)	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	0 (0.0)
	Pueblo of Santo Domingo	1 (0.7)	3 (2.2)	0 (0.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Pueblo of Taos	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Quinault	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Red Lake Band of Chippewa	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Seneca Nation of New York	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Suquamish	1 (0.7)	0 (0.0)	1 (1.0)	0 (0.0)	1 (0.8)	0 (0.0)	0 (0.0)
	Turtle Mountain Band of Chippewa	2 (1.1)	3 (2.2)	1 (1.2)	0 (0.0)	2 (1.3)	0 (0.0)	0 (0.0)
	White Mountain Apache	2 (1.2)	0 (0.0)	3 (1.7)	0 (0.0)	2 (1.4)	0 (0.0)	0 (0.0)
	Zuni	3 (1.4)	0 (0.0)	4 (2.0)	0 (0.0)	3 (1.7)	0 (0.0)	0 (0.0)
	Inuit Eskimo	2 (0.9)	0 (0.0)	2 (1.3)	0 (0.0)	2 (1.1)	0 (0.0)	0 (0.0)
	Saponi	1 (1.0)	4 (3.3)	0 (0.0)	14 (11.6)	0 (0.0)	0 (0.0)	0 (0.0)
	Taino	1 (0.9)	4 (3.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	23 (17.2)
	Seminole (Unspecified)	2 (1.2)	4 (3.3)	1 (1.0)	14 (11.6)	1 (0.8)	0 (0.0)	0 (0.0)
	Asa'carsarmiut	1 (0.8)	3 (2.7)	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	0 (0.0)
	Cherikawa	1 (0.7)	0 (0.0)	1 (1.1)	0 (0.0)	0 (0.0)	0 (0.0)	20 (15.0)
	Kahnawake Mohawk	1 (0.8)	3 (2.7)	0 (0.0)	0 (0.0)	1 (0.9)	0 (0.0)	0 (0.0)
	Kickapoo (unspecified)	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Me-Wuk (unspecified)	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	Black Creek	1 (0.8)	0 (0.0)	1 (1.1)	11 (9.3)	0 (0.0)	0 (0.0)	0 (0.0)
	Algaaciq	1 (0.9)	0 (0.0)	1 (1.2)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)

Chicano descent, 8 percent are Puerto Rican, and 5 percent are Cuban. The ethnic distribution of the non-recipient Hispanic Americans is in good agreement with that of the Scholars.

Finally, close to a third of the Pacific Islanders in the Asian/Pacific Islander subpopulation of non-recipients are Hawaiian, 21 percent are Guamanian or Chamorro, and 21 percent Fijian.[14] The remaining non-recipients are distributed in small numbers across other islands.

Tribe. Non-recipients in the American Indian/Alaska Native subpopulation are members of more than 40 different state and federally recognized American Indian and Alaska Native tribes. About 19 percent belong to the Cherokee (unspecified tribe) and 18 percent to the Navajo. The remaining non-recipients are distributed in small numbers across the other tribes listed in Exhibit 10.

2.2.2 Geographic Background

As shown in Exhibit 11, about 1 percent of the non-recipients completed high school outside of the United States in one of the U.S. possessions or in a foreign country. As was the case for the Scholars, close to two-thirds of the non-recipients completed secondary school in the Pacific, South Atlantic, or West South Central divisions of the country. Less than 10 percent of the non-recipients completed high school in each of the six remaining divisions. The distributions of male and female non-recipients across the divisions are similar. However, there are larger shares of males than females in the West South Central states and larger shares of females than males in the New England and Midwest states.

[14] Pacific Islander non-recipients include all non-recipients in the Asian/Pacific Islander subpopulation of non-recipients who identified themselves as Pacific Islanders in the survey.

Exhibit 11. Geographic Background of the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Census Division of HS	New England	1 (0.3)	0 (0.4)	2 (0.4)	2 (0.7)	0 (0.0)	2 (0.8)	1 (0.4)
	Middle Atlantic	8 (0.7)	8 (1.5)	7 (0.8)	9 (1.5)	1 (0.8)	9 (1.6)	7 (1.2)
	Midwest	9 (0.8)	6 (1.3)	10 (0.9)	15 (1.8)	0 (0.0)	8 (1.6)	6 (1.1)
	West North Central	5 (0.6)	6 (1.3)	5 (0.7)	5 (1.2)	9 (2.5)	7 (1.4)	4 (0.9)
	South Atlantic	17 (1.0)	15 (2.0)	18 (1.2)	27 (2.3)	4 (1.8)	13 (1.9)	15 (1.6)
	East South Central	5 (0.6)	4 (1.2)	6 (0.7)	14 (1.8)	1 (0.7)	1 (0.6)	1 (0.5)
	West South Central	23 (1.2)	27 (2.4)	21 (1.3)	20 (2.1)	39 (4.2)	9 (1.6)	28 (2.0)
	Mountain	9 (0.8)	10 (1.5)	9 (0.9)	1 (0.4)	32 (4.0)	6 (1.3)	12 (1.5)
	Pacific	22 (1.1)	23 (2.2)	21 (1.2)	7 (1.4)	13 (2.9)	45 (2.8)	25 (1.9)
	Outside the U.S.	1 (0.2)	0 (0.3)	1 (0.3)	0 (0.0)	0 (0.0)	1 (0.6)	1 (0.5)

2.2.3 Socio-Economic Background

Educational attainment of the parents. As shown in Exhibit 12, at least 79 percent of the non-recipients came from families where either their father or mother had completed at least a high school degree. Twenty-four percent of their fathers and 25 percent of their mothers went on to earn at least a bachelor's degree, and 10 percent of their fathers and 9 percent of their mothers obtained advanced degrees.[15]

As was the case for the Scholars, the educational attainment levels of the parents of the non-recipients vary by minority group. Like the Scholars, the parents of the Hispanic American non-recipients show lower levels of educational attainment compared to the parents of the other minority groups. Most of the fathers of the African American (91 percent) and American Indian/Alaska Native (92 percent) non-recipients completed at least a high school degree. Both were more likely to complete high school than the fathers of the Asian/Pacific Islanders (78 percent). At 67 percent, the fathers of the Hispanic American non-recipients were less likely than the other groups to have completed high school. The high school completion rates of their mothers follow a similar but not identical pattern. Relatively more of the mothers of African American (94 percent) and American Indian/Alaska Native (98 percent) non-recipients completed at least a high school degree compared to the mothers of Asian/Pacific Islander (70 percent) and Hispanic American (70 percent) non-recipients.

A comparison of the fathers shows that the fathers of the Asian/Pacific Islanders (33 percent) were more likely than the fathers of African American (23 percent) and Hispanic American (18 percent) non-recipients to have obtained a bachelor's degree. A comparison of the rates at which mothers of the non-recipients attained at least a bachelor's degree shows that the mothers of Hispanic Americans (20 percent) were less likely than the mothers of African American non-recipients (31 percent) to obtain a bachelor's degree.

The fathers of Asian/Pacific Islanders (16 percent) obtained advanced degrees at higher rates than did the fathers of both African American (9 percent) and Hispanic American (9 percent) non-recipients. However, the mothers of the non-recipients did not differ significantly in how likely they were to have obtained an advanced degree.

As a whole, the parents of the non-recipients were more likely than their Scholar counterparts to earn high school diplomas. The fathers of the Hispanic American non-recipients were more likely to have completed high school compared to the fathers of the Hispanic American Scholars. The mothers of the African American, Asian/Pacific Islander, and Hispanic American non-recipients were more likely to have completed high school than their Scholar counterparts.

[15] To obtain the percent completing at least a high school degree, we took the compliment of the percent with less than a high school degree (100 minus the percent with less than a high school degree). We computed the percent completing at least a bachelor's degree by summing the percents earning a bachelor's, master's, or doctoral degree as their highest degree. Similarly, we obtained the percent with advanced degrees by summing the percents with a master's or doctoral degree as their highest degree. Note that the summed percents may not agree with those based on the percents in the exhibits due to rounding error.

Exhibit 12. Educational Attainment of the Cohort V Non-Recipients' Parents for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Father's highest level of educational attainment	Less than high school	21 (1.1)	20 (2.2)	21 (1.3)	9 (1.5)	8 (2.4)	22 (2.4)	33 (2.1)
	GED	3 (0.5)	3 (1.0)	3 (0.5)	2 (0.8)	5 (1.9)	0 (0.0)	4 (0.9)
	High school graduation	27 (1.2)	28 (2.5)	27 (1.4)	34 (2.4)	26 (3.9)	22 (2.4)	25 (2.0)
	Some college	25 (1.2)	25 (2.4)	25 (1.4)	32 (2.4)	33 (4.0)	23 (2.5)	20 (1.8)
	Bachelor's degree	13 (0.9)	15 (2.0)	13 (1.1)	14 (1.8)	18 (3.3)	17 (2.2)	10 (1.3)
	Master's degree or equivalent	7 (0.7)	6 (1.4)	7 (0.8)	7 (1.4)	7 (2.3)	10 (1.7)	6 (1.0)
	Doctorate	3 (0.5)	3 (0.9)	3 (0.6)	2 (0.8)	2 (1.3)	6 (1.3)	3 (0.8)
Mother's highest level of educational attainment	Less than high school	20 (1.0)	22 (2.3)	19 (1.1)	6 (1.3)	2 (1.2)	30 (2.7)	30 (2.1)
	GED	4 (0.5)	2 (0.7)	4 (0.7)	3 (0.8)	4 (1.9)	2 (0.9)	5 (0.9)
	High school graduation	21 (1.1)	26 (2.4)	19 (1.3)	18 (2.0)	21 (3.5)	24 (2.5)	21 (1.9)
	Some college	31 (1.3)	28 (2.5)	32 (1.5)	42 (2.5)	44 (4.3)	18 (2.2)	24 (1.9)
	Bachelor's degree	16 (1.0)	17 (2.1)	15 (1.2)	21 (2.1)	15 (3.1)	14 (2.1)	13 (1.5)
	Master's degree or equivalent	8 (0.8)	6 (1.3)	9 (0.9)	10 (1.5)	13 (2.9)	9 (1.6)	6 (1.1)
	Doctorate	1 (0.3)	1 (0.4)	1 (0.3)	1 (0.4)	0 (0.0)	2 (0.8)	1 (0.4)
Father's attainment (HS degree)	At least a HS degree	79 (1.1)	80 (2.2)	79 (1.3)	91 (1.5)	92 (2.4)	78 (2.4)	67 (2.1)
	Less than a HS degree	21 (1.1)	20 (2.2)	21 (1.3)	9 (1.5)	8 (2.4)	22 (2.4)	33 (2.1)
Father's attainment (bachelor's degree)	At least a bachelor's degree	24 (1.2)	24 (2.4)	24 (1.3)	23 (2.2)	28 (3.8)	33 (2.7)	18 (1.7)
	Less than a bachelor's degree	76 (1.2)	76 (2.4)	76 (1.3)	77 (2.2)	72 (3.8)	67 (2.7)	82 (1.7)
Father's attainment (advanced degree)	Advanced degree	10 (0.8)	10 (1.6)	11 (1.0)	9 (1.5)	10 (2.6)	16 (2.1)	9 (1.2)
	Less than an advanced degree	90 (0.8)	90 (1.6)	89 (1.0)	91 (1.5)	90 (2.6)	84 (2.1)	91 (1.2)
Mother's attainment (HS degree)	At least a HS degree	80 (1.0)	78 (2.3)	81 (1.1)	94 (1.3)	98 (1.2)	70 (2.7)	70 (2.1)
	Less than a HS degree	20 (1.0)	22 (2.3)	19 (1.1)	6 (1.3)	2 (1.2)	30 (2.7)	30 (2.1)
Mother's attainment (bachelor's degree)	At least a bachelor's degree	25 (1.2)	23 (2.4)	26 (1.4)	31 (2.4)	28 (3.9)	25 (2.5)	20 (1.8)
	Less than a bachelor's degree	75 (1.2)	77 (2.4)	74 (1.4)	69 (2.4)	72 (3.9)	75 (2.5)	80 (1.8)
Mother's attainment (advanced degree)	Advanced degree	9 (0.8)	6 (1.3)	11 (1.0)	10 (1.5)	13 (2.9)	11 (1.8)	7 (1.1)
	Less than an advanced degree	91 (0.8)	94 (1.3)	89 (1.0)	90 (1.5)	87 (2.9)	89 (1.8)	93 (1.1)

Some differences in educational attainment are apparent in the rates at which parents of Scholars and non-recipients earn bachelor's and advanced degrees. The fathers of the non-recipients were more likely to have completed at least a bachelor's degree compared to the fathers of the Scholars. However, the rates at which the Scholars' and non-recipients' fathers attained advanced degrees and mothers attained at least bachelor's or advanced degrees were close in value. Comparing the Scholars and non-recipients by minority group, we found that, for the American Indians/Alaska Natives, fathers of the non-recipients were more likely to have attained at least a bachelor's degree (28 percent) compared to the fathers of the Scholars (16 percent). In addition, fathers of Hispanic American non-recipients were more likely to have attained an advanced degree (9 percent) compared to the fathers of their Scholar counterparts (5 percent).

2.2.4 Educational Background and Secondary School Experiences

The baseline survey also asked the non-recipients several questions regarding their educational background and secondary school experiences. Their answers to this set of questions are shown in Exhibit 13.

Exhibit 13. Educational Background and Secondary School Experiences of the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Race majority of the students in nominee's HS classes	Strongly Disagree	31 (1.3)	29 (2.5)	32 (1.4)	36 (2.5)	17 (3.1)	45 (2.9)	24 (1.9)
	Disagree	32 (1.3)	32 (2.6)	33 (1.5)	30 (2.4)	19 (3.3)	38 (2.8)	35 (2.2)
	Agree	17 (1.0)	19 (2.2)	16 (1.1)	13 (1.8)	34 (4.1)	12 (1.9)	17 (1.7)
	Strongly Agree	20 (1.1)	20 (2.3)	20 (1.3)	20 (2.1)	31 (4.0)	5 (1.3)	24 (1.9)
No. of AP exams taken in high school	None	29 (1.2)	29 (2.5)	28 (1.4)	40 (2.6)	42 (4.3)	20 (2.3)	20 (1.8)
	One	17 (1.0)	17 (2.1)	17 (1.2)	19 (2.0)	23 (3.6)	14 (2.0)	15 (1.6)
	Two	17 (1.0)	18 (2.2)	16 (1.2)	16 (1.9)	14 (3.1)	15 (2.1)	19 (1.8)
	Three	13 (0.9)	10 (1.7)	15 (1.1)	12 (1.6)	10 (2.7)	12 (1.9)	16 (1.6)
	Four or more	25 (1.2)	26 (2.3)	24 (1.3)	13 (1.8)	10 (2.4)	39 (2.8)	31 (2.1)
Attended a religious high school	Yes	7 (0.7)	9 (1.6)	6 (0.8)	6 (1.3)	9 (2.4)	5 (1.3)	8 (1.2)
	No	93 (0.7)	91 (1.6)	94 (0.8)	94 (1.3)	91 (2.4)	95 (1.3)	92 (1.2)
Attended a private high school	Yes	8 (0.8)	9 (1.6)	8 (0.9)	7 (1.3)	12 (2.9)	7 (1.5)	8 (1.2)
	No	92 (0.8)	91 (1.6)	92 (0.9)	93 (1.3)	88 (2.9)	93 (1.5)	92 (1.2)
No. of years of math completed in HS	Two	1 (0.3)	1 (0.6)	1 (0.4)	1 (0.6)	1 (0.7)	1 (0.7)	1 (0.4)
	Three	16 (1.0)	16 (2.0)	16 (1.2)	15 (1.8)	27 (3.9)	14 (2.0)	16 (1.6)
	Four or more	82 (1.0)	83 (2.1)	82 (1.2)	84 (1.9)	72 (3.9)	84 (2.1)	83 (1.7)
No. of years of science completed in HS	One	0 (0.1)	0 (0.0)	0 (0.1)	0 (0.0)	0 (0.0)	0 (0.4)	0 (0.0)
	Two	4 (0.5)	5 (1.2)	3 (0.6)	2 (0.7)	8 (2.4)	2 (0.9)	5 (1.0)
	Three	30 (1.3)	28 (2.5)	32 (1.5)	27 (2.3)	31 (4.0)	21 (2.3)	38 (2.2)
	Four or more	66 (1.3)	68 (2.6)	65 (1.5)	71 (2.4)	61 (4.2)	76 (2.4)	57 (2.2)

Race majority of students in high school classes. About 63 percent of the non-recipients disagreed or strongly disagreed that the classes they took in high school were with students mostly of their race—indicating that, like the Scholars, most non-recipients were probably in classes where they were not the majority race.

As a whole, Asian/Pacific Islander non-recipients were less likely than the other minority groups to be in classes where they were the majority race. With only 5 percent of the Asian/Pacific Islanders strongly agreeing that they were in the majority race, this group was less likely than the other three groups to be in classes where they were the majority race. In comparison, 20 percent of the African American, 31 percent of the American Indian/Alaska Native, and 24 percent of the Hispanic American non-recipients strongly agreed that they were in the majority race in their high school classes.

Asian/Pacific Islander and African American non-recipients were more likely to strongly disagree that they were in classes where they were the majority race (45 percent and 36 percent, respectively) compared to Hispanic Americans (24 percent) and American Indians (17 percent).

At the level of the populations, the Scholars were more likely than were the non-recipients to strongly agree that they were in the race majority in their high school classes (25 percent vs. 20 percent, respectively). Further, non-recipients were more likely than Scholars to disagree that they were in the race majority in their high school classes (32 percent vs. 25 percent, respectively). This suggests that the non-recipients were in somewhat more racially diverse classrooms than were the Scholars.

Attendance at religious or private high schools. The large majority of non-recipients attended public high schools. Eight percent were enrolled in private schools, and 7 percent in religious schools. There were no significant differences among minority groups in their attendance levels at religious or private high schools.

The Scholars and non-recipients were similar in the rates at which they attended religious high schools, but the Scholars were more likely to attend private high schools.

Advanced placement exams. Seventy-one percent of the non-recipients took at least one advanced placement exam while in high school.[16] About 55 percent took two or more advanced placement exams, 38 percent, three or more, and 25 percent, four or more while in high school.

Eighty percent of both Asian/Pacific Islander and Hispanic American non-recipients took at least one advanced placement exam. These groups were much more likely than the African Americans and American Indians/Alaska Natives to do so. Forty percent of African American and 42 percent of American Indian/Alaska Native non-recipients did not take any advanced placement exams during their high school careers.

Asian/Pacific Islander and Hispanic American non-recipients were more likely than the other minority groups to complete four or more advanced placement exams. Thirty-nine percent of Asian/Pacific Islanders

[16] To find the percent who took at least one advanced placement exam, we took the compliment of the percent who took none (100 minus the percent who took none). To find the percent who took two or more exams, we summed the percents who took two, three, or four or more exams. Similarly, to find the percent who took three or more exams, we added the percents who took three or four or more exams.

and 31 percent of Hispanic Americans reported that they took four or more advanced placement exams in high school compared to 13 and 10 percent of African American and American Indian/Alaska Native non-recipients, respectively.

Overall, the Scholars took more advanced placement exams than the non-recipients. The non-recipients were more likely to have taken no advanced placement exams and less likely to have taken four or more advanced placement exams, compared to the Scholars. With the exception of the American Indians/Alaska Natives, for each minority group the non-recipients also took fewer advanced placement exams than did their respective Scholar counterparts. For African Americans and Hispanic Americans, the non-recipient minority groups were more likely to have taken no advanced placement exams and less likely to have completed four or more advanced placement exams compared to the Scholars. For Asian/Pacific Islanders, although the Scholars were more likely than non-recipients to take four or more advanced placement exams, the difference in percent taking no advanced placement exams was not statistically significant.

Years of math and science in high school. In terms of the number of years taken, the non-recipients, like the Scholars, have relatively strong backgrounds in math and science. About 98 percent completed at least three years of math while in high school.[17] Eighty-two percent took four or more years of math. Ninety-six percent reported taking three or more years of science in high school, and 66 percent said they took four or more years of science.

Like the Scholars, male and female non-recipients did not differ significantly in their tendency to take four or more years of science (68 percent vs. 65 percent, respectively). However, unlike the case with the Scholars, female non-recipients did not differ significantly from the males in the percent taking four or more years of math (82 percent vs. 83 percent, respectively).

The math backgrounds of the non-recipients vary somewhat by minority group. American Indian/Alaska Native non-recipients (72 percent) were less likely than both African American (84 percent) and Asian/Pacific Islanders (84 percent) to take four or more years of math while in high school. Although the share of Hispanic Americans taking four or more years of math, at 83 percent, was nearly the same as shares of African Americans and Asian/Pacific Islanders, the difference between Hispanic American and American Indian/Alaska Native non-recipients was not significant.

Asian/Pacific Islanders (76 percent) were more likely than American Indians/Alaska Natives (61 percent) and Hispanic Americans (57 percent) to take four or more years of science. African Americans (71 percent) were not significantly different from Asian/Pacific Islanders and American Indians/Alaska Natives in their likelihood to take four or more years of science but more likely than Hispanic Americans to do so.

At the level of the population, the Scholars were more likely than the non-recipients to have taken four or more years of math and science in high school. With regard to the percentages taking four or more years of math or science, Asian/Pacific Islander Scholars took more math than their non-recipient counterparts;

[17] To find the percent of non-recipients who took three or more years of math or science, we summed the percents who took three or four or more years of math or science.

Hispanic American Scholars took more science than their non-recipient counterparts. At the level of minority group, there were no other statistically significant differences between the shares of Scholars and non-recipients taking four or more years of math or science.

2.2.5 Educational Aspirations and Expectations

As part of the baseline survey, the non-recipients were asked to indicate the highest degree they expected to earn. Exhibit 14 displays the answers they gave. At the time of the baseline survey, 99 percent of the non-recipients expected to complete at least a four-year bachelor's degree, and 82 percent predicted that they would go on to complete an advanced degree.[18] As was the case for the Scholars, American Indian/Alaska Native non-recipients were less likely than the other minority groups to predict that they would complete an advanced degree. Only 60 percent of the American Indians/Alaska Natives believed they would earn at least a master's degree compared to 91, 82, and 81 percent of the African American, Asian/Pacific Islander, and Hispanic American non-recipients, respectively. African Americans were the most likely of all minority groups to think that they would earn at least a master's degree. However, African American and Asian/Pacific Islander non-recipients did not differ significantly in the percent who believed that they would earn a first professional or doctoral degree (55 percent for both). Both were more likely than American Indian/Alaska Native (31 percent) and Hispanic American (43 percent) non-recipients to anticipate that they would earn a professional or doctoral degree.

Overall, the educational aspirations of the Scholars were higher than those of the non-recipients. At the level of at least the bachelor's, master's, and first professional or doctoral degree, the Scholars were more likely than the non-recipients to believe that they would earn at least that degree. With respect to the four minority groups, the percentages of Scholars and non-recipients who expected to earn at least a bachelor's degree were close in value. However, for every minority group, the Scholars were more likely than their non-recipient counterparts to believe they would earn at least a master's or a first professional or doctoral degree.

[18] To find the percent expecting to complete at least a bachelor's degree, we summed the percents expecting to complete a post-baccalaureate certificate or a bachelor's, master's, first professional, or doctoral degree as their highest level of educational attainment. The percent aspiring to obtain an advanced degree was computed by summing the percents expecting to complete a master's, first professional, or doctoral degree as their highest degree. Note that the summed percents may not agree with those based on the percents in the exhibits due to rounding error.

Exhibit 14. Educational Aspirations of the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Educational aspirations (highest level of aspiration)	Less than two years of college	0 (0.1)	0 (0.0)	0 (0.2)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.3)
	Two or more years of college	1 (0.2)	0 (0.4)	1 (0.3)	0 (0.3)	4 (1.7)	0 (0.4)	0 (0.3)
	Bachelor’s degree	16 (1.0)	14 (1.9)	16 (1.2)	7 (1.4)	34 (4.3)	16 (2.1)	17 (1.8)
	Post-baccalaureate certificate	1 (0.3)	1 (0.7)	1 (0.3)	1 (0.6)	2 (1.2)	1 (0.7)	1 (0.4)
	Master’s degree (M.A., M.S., M.B.A., etc.)	34 (1.4)	40 (2.8)	32 (1.5)	35 (2.5)	29 (4.1)	28 (2.6)	38 (2.2)
	First professional degree (M.D., J.D., D.D.S., O.D.)	19 (1.1)	17 (2.2)	19 (1.3)	22 (2.2)	12 (3.0)	24 (2.5)	15 (1.6)
	Doctoral degree (PH.D., ED.D., D.P.H., etc.)	29 (1.3)	26 (2.5)	31 (1.5)	33 (2.5)	19 (3.6)	31 (2.7)	28 (2.1)
Educational aspirations (bachelor's degree)	At least a bachelor's degree	99 (0.3)	100 (0.4)	99 (0.3)	100 (0.3)	96 (1.7)	100 (0.4)	99 (0.4)
	Less than a bachelor's degree	1 (0.3)	0 (0.4)	1 (0.3)	0 (0.3)	4 (1.7)	0 (0.4)	1 (0.4)
Educational aspirations (master's degree)	At least a master's degree	82 (1.1)	84 (2.0)	82 (1.2)	91 (1.5)	60 (4.4)	82 (2.2)	81 (1.8)
	Less than a master's degree	18 (1.1)	16 (2.0)	18 (1.2)	9 (1.5)	40 (4.4)	18 (2.2)	19 (1.8)
Educational aspirations (first professional or doctoral degree)	First professional or doctoral degree	48 (1.4)	44 (2.8)	50 (1.6)	55 (2.6)	31 (4.2)	55 (2.9)	43 (2.3)
	Less than a professional or doctoral degree	52 (1.4)	56 (2.8)	50 (1.6)	45 (2.6)	69 (4.2)	45 (2.9)	57 (2.3)

PART III: COLLEGE-GOING RATES AND COLLEGE TYPES OF THE SCHOLARS AND NON-RECIPIENTS

To assess the college-going rates of the Scholars and non-recipients, they were asked about their current enrollment status at the time of the baseline survey. The survey asked them to report whether they were currently enrolled in college.

As part of the baseline survey, Scholars who were ever enrolled in a college or university were asked to supply the name of the postsecondary institution of their current or last enrollment. The names of the institutions were matched with their codes in the Integrated Postsecondary Education Data System (IPEDS) to obtain background information on the Scholars' colleges from the National Center of Education Statistics (NCES) IPEDS website.

Exhibit 15 displays the background characteristics of the Scholars' postsecondary institutions for the population as a whole, by gender, and by minority group.

3.1 College-Going Rates and College Types of the Scholars

Enrollment rates. Exhibit 15 shows that nearly all (rounded up to 100 percent in the exhibit) of the Cohort V Scholars were enrolled in college at the time they completed the baseline survey, between June 16 and October 3 of 2005, roughly one year after they graduated from high school. The college-going rates for all four minority groups were high.

Level of institution. For the overwhelming majority of Scholars—about 98 percent—the institution at which they were currently or last enrolled offered at least a bachelor's degree. The remaining 2 percent of Scholars attended junior colleges. Compared to the other minority groups, American Indians/Alaska Natives were less likely to enroll in institutions offering at least a bachelor's degree (see Exhibit 15).

Control of institution. More than half of Scholars (55 percent) reported that the institution of their current or last enrollment was a publicly controlled institution. Females were more likely than males to attend a public institution, and males were more likely than females to attend a private, not-for-profit institution. At least half of the Scholars in each minority group were enrolled in public institutions; some of the differences by minority group were significant. American Indians/Alaska Natives (69 percent) were more likely than the other groups to be enrolled in a public institution. Hispanic Americans (55 percent) were more likely than Asian/Pacific Islanders (50 percent) to be enrolled in a public institution. Hispanic Americans and African Americans (52 percent) were similar in the rates at which they attended public institutions. African Americans and Asian/Pacific Islanders were also similar in their rates of attending public institutions.

Exhibit 15. Enrollment Status and Background Characteristics of the Institutions of Enrollment for the Population of Cohort V Scholars as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
College enrollment status	Yes	100 (0.1)	100 (0.1)	100 (0.1)	100 (0.1)	99 (0.3)	100 (0.0)	100 (0.0)
	No	0 (0.1)	0 (0.1)	0 (0.1)	0 (0.1)	1 (0.3)	0 (0.0)	0 (0.0)
Level of institution	Four or more years	98 (0.2)	98 (0.3)	98 (0.2)	99 (0.2)	92 (1.0)	99 (0.2)	99 (0.2)
	At least 2 but less than 4 years	2 (0.2)	2 (0.3)	2 (0.2)	1 (0.2)	8 (1.0)	1 (0.2)	1 (0.2)
Control of institution	Public	55 (0.6)	51 (1.1)	57 (0.6)	52 (1.0)	69 (2.0)	50 (1.2)	55 (0.8)
	Private not-for-profit	45 (0.6)	49 (1.1)	43 (0.6)	48 (1.0)	31 (2.0)	50 (1.2)	45 (0.8)
	Private for-profit	0 (0.1)	0 (0.1)	0 (0.1)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.1)
Highest degree offered	Doctoral	14 (0.4)	12 (0.7)	15 (0.5)	14 (0.7)	8 (1.3)	10 (0.7)	18 (0.6)
	Doctoral and first professional	62 (0.6)	64 (1.1)	61 (0.6)	60 (1.0)	50 (2.1)	77 (1.0)	63 (0.8)
	Master’s	13 (0.4)	12 (0.7)	13 (0.5)	12 (0.7)	21 (1.6)	8 (0.7)	11 (0.5)
	Master’s and first professional	2 (0.2)	2 (0.3)	3 (0.2)	4 (0.4)	6 (0.9)	0 (0.0)	1 (0.1)
	Bachelor’s	7 (0.3)	9 (0.6)	6 (0.3)	9 (0.6)	7 (1.1)	4 (0.5)	6 (0.4)
	Associate’s	2 (0.2)	2 (0.3)	2 (0.2)	1 (0.2)	8 (1.0)	1 (0.2)	1 (0.2)
Historically black college or university	Yes	5 (0.3)	6 (0.4)	5 (0.3)	16 (0.7)	0 (0.0)	0 (0.0)	0 (0.0)
	No	95 (0.3)	94 (0.4)	95 (0.3)	84 (0.7)	100 (0.0)	100 (0.0)	100 (0.0)
Tribal college or university	Yes	1 (0.1)	0 (0.0)	1 (0.1)	0 (0.0)	4 (0.6)	0 (0.0)	0 (0.0)
	No	99 (0.1)	100 (0.0)	99 (0.1)	100 (0.0)	96 (0.6)	100 (0.0)	100 (0.0)

Highest degree offered. About 62 percent of Scholars, including 64 percent of males and 61 percent of females, were currently attending or were last enrolled in colleges or universities offering both doctoral and first professional programs as their highest degree-granting programs. The large majority of Asian/Pacific Islanders (77 percent) went to institutions of this type, surpassing the shares of the other three minority groups. American Indians/Alaska Natives, at 50 percent, were the least likely of any minority group to attend an institution offering both doctoral and first professional programs.

Historically black college or university.[19] Five percent of Scholars were either currently or last enrolled in historically black colleges or universities. Although 16 percent of African American Scholars attended historically black colleges or universities, no Scholars from the other three groups did so.

[19] “The Higher Education Act of 1965, as amended, defines an HBCU as: ‘...any historically black college or university that was established prior to 1964, whose principal mission was, and is, the education of black Americans, and that is accredited by a nationally recognized accrediting agency or association determined by the Secretary [of Education] to be a reliable authority as to the quality of training offered or is, according to such an agency or association, making reasonable progress toward accreditation’” (National Center for Education Statistics, “IPEDS Glossary,” http://nces.ed.gov/ipeds/glossary/pdf/IPEDSglossary.pdf [accessed May 4, 2007]).

Tribal college or university.[20] Only 1 percent of all Scholars reported that they were currently or last enrolled in a tribal college or university. Consisting entirely of female American Indians/Alaska Natives, this group of Scholars accounts for 4 percent of the American Indian/Alaska Native population of Scholars.

3.2 College-Going Rates and College Types of the Non-Recipients

Enrollment rates. The enrollment rates for the non-recipients are shown in Exhibit 16. We see in this exhibit that about 98 percent of non-recipients were enrolled in college or university at the time of the baseline survey, roughly one year after their graduation from high school. Although high, this rate is significantly below that of the Scholars.

Exhibit 16. Enrollment Status and Background Characteristics of the Institutions of Enrollment for the Population of Cohort V Non-Recipients as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
College enrollment status	Yes	98 (0.4)	98 (0.7)	98 (0.5)	99 (0.6)	93 (2.3)	98 (0.8)	98 (0.6)
	No	2 (0.4)	2 (0.7)	2 (0.5)	1 (0.6)	7 (2.3)	2 (0.8)	2 (0.6)
Level of institution	Four or more years	93 (0.7)	93 (1.3)	93 (0.8)	97 (0.9)	80 (3.5)	95 (1.2)	92 (1.2)
	At least 2 but less than 4 years	7 (0.7)	7 (1.3)	7 (0.8)	3 (0.9)	20 (3.5)	5 (1.2)	8 (1.2)
Control of institution	Public	68 (1.3)	66 (2.7)	69 (1.4)	63 (2.5)	89 (2.6)	67 (2.7)	67 (2.1)
	Private not-for-profit	31 (1.3)	33 (2.7)	30 (1.4)	36 (2.5)	9 (2.4)	33 (2.7)	32 (2.1)
	Private for-profit	1 (0.3)	1 (0.6)	1 (0.3)	1 (0.5)	2 (1.1)	0 (0.0)	1 (0.4)
Highest degree offered	Doctoral	18 (1.1)	18 (2.1)	18 (1.2)	22 (2.1)	7 (2.3)	10 (1.7)	21 (1.8)
	Doctoral and first professional	53 (1.4)	49 (2.7)	54 (1.6)	46 (2.5)	44 (4.3)	69 (2.6)	52 (2.3)
	Master's	15 (1.0)	19 (2.2)	13 (1.1)	18 (2.0)	15 (3.0)	12 (1.9)	14 (1.6)
	Master's and first professional	2 (0.4)	2 (0.9)	2 (0.4)	3 (0.9)	6 (2.2)	2 (0.7)	1 (0.3)
	Bachelor's	5 (0.6)	6 (1.4)	5 (0.7)	8 (1.4)	7 (2.3)	3 (1.0)	4 (0.9)
	Associate's	7 (0.7)	7 (1.3)	7 (0.8)	3 (0.9)	20 (3.5)	5 (1.2)	8 (1.2)
Historically black college or university	Yes	6 (0.6)	4 (1.2)	7 (0.8)	18 (2.0)	0 (0.0)	0 (0.0)	0 (0.3)
	No	94 (0.6)	96 (1.2)	93 (0.8)	82 (2.0)	100 (0.0)	100 (0.0)	100 (0.3)
Tribal college or university	Yes	0 (0.1)	0 (0.0)	0 (0.1)	0 (0.0)	1 (1.0)	0 (0.0)	0 (0.0)
	No	100 (0.1)	100 (0.0)	100 (0.1)	100 (0.0)	99 (1.0)	100 (0.0)	100 (0.0)

[20] A tribal college or university is "an institutional classification developed by the Andrew W. Carnegie Foundation for the Advancement of Teaching. Tribal Colleges and Universities, with few exceptions, are tribally controlled and located on reservations. They are all members of the American Indian Higher Education Consortium" (National Center for Education Statistics, "IPEDS Glossary," http://nces.ed.gov/ipeds/glossary/pdf/IPEDSglossary.pdf [accessed May 4, 2007]).

In terms of the four minority groups, the enrollment rate for American Indian/Alaska Native non-recipients (93 percent) was slightly below the rate of the African American, Asian/Pacific Islander, and the Hispanic American non-recipients, but not statistically different. Comparing the enrollment rate for Scholars and non-recipients by minority group, we found that American Indian/Alaska Native Scholars were more likely to be enrolled than their non-recipient counterparts, but there were no significant differences for the other minority groups.

Level of institution. For about 93 percent of non-recipients, the institution in which they were currently or last enrolled offered at least a bachelor's degree (see Exhibit 16). African Americans, Asian/Pacific Islanders, and Hispanic Americans were more likely than American Indians/Alaska Natives to attend an institution offering at least the bachelor's degree. African Americans were also more likely than Hispanic Americans to attend this type of institution. There were no other statistically significant differences by minority group or by gender.

Compared to the Scholars, the non-recipients, as a whole, by gender, and for three minority groups, were less likely to have enrolled in at least a four-year institution. Although the percent of African Americans who had enrolled in four-year institutions is also lower for the non-recipients than for the Scholars, the difference was not statistically significant.

Control of institution. Sixty-eight percent of non-recipients reported that the institution in which they were currently or last enrolled was a public postsecondary institution, with no statistically significant differences by gender. With 89 percent enrolled in public institutions, American Indians/Alaska Natives were more likely than African Americans, Asian/Pacific Islanders, and Hispanic Americans to go to public institutions.

Compared to the Scholars, non-recipients were much less likely to go to private, not-for-profit colleges and universities. This pattern holds for males and females, as well as for the four minority groups.

Highest degree offered. A little over half of non-recipients were currently or last enrolled in colleges or universities offering both doctoral and first professional degrees as their highest degrees, a proportion below that of the Scholars. Non-recipients were also less likely than Scholars to have enrolled in institutions offering the bachelor's or the doctoral and first professional as the highest degrees. However, non-recipients were more likely than Scholars to attend an institution offering the doctoral (without professional degrees) as the highest degree, or an institution offering the associate's as the highest degree. With respect to individual minority groups, American Indian/Alaska Native, Asian/Pacific Islander, and Hispanic American non-recipients were more likely than were the Scholars in those groups to attend institutions offering the associate's degree as the highest degree. For the African Americans, Asian/Pacific Islanders, and Hispanic Americans, non-recipients were less likely than Scholars to attend institutions offering both doctoral and first professional degrees.

Historically black college or university. Six percent of non-recipients were currently or last enrolled in historically black colleges or universities at the time of the survey, a proportion that does not differ significantly from that of the Scholars. Eighteen percent of African American non-recipients attended a historically black college or university.

Tribal college or university. Less than 1 percent of non-recipients were currently or last enrolled in tribal colleges or universities at the time of the survey. As was the case for Scholars, the enrollees were all female American Indians/Alaska Natives.

PART IV: EARLY OUTCOMES AND EXPERIENCES OF THE COHORT V SCHOLARS AND NON-RECIPIENTS

To assess the impact of the scholarship program on the early outcomes and experiences of the Scholars while in college, the Scholars and non-recipients were asked a series of questions regarding their work experiences, reasons for working, and finances. This section presents the responses they supplied to this set of questions in the baseline survey—roughly one year after they graduated from high school—for the populations as a whole and by gender and minority group.

4.1 Early Outcomes and Experiences of the Scholars

4.1.1 Work for Pay

As part of the baseline survey, Scholars who were currently enrolled in college were asked a series of questions regarding their work activities, including whether they worked, how many hours and at what rate they worked for pay while in school, and their reasons for working. Exhibits 17 and 18 show their responses to this set of questions.

Exhibit 17. Work Activities of the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) or Mean (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent or Mean (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Currently work f or pay	Yes	52 (0.6)	50 (1.1)	53 (0.7)	56 (1.0)	56 (2.1)	38 (1.2)	52 (0.8)
	No	48 (0.6)	50 (1.1)	47 (0.7)	44 (1.0)	44 (2.1)	62 (1.2)	48 (0.8)
Work on or off campus	On campus	26 (0.7)	29 (1.3)	24 (0.8)	28 (1.2)	11 (1.8)	31 (1.8)	29 (1.0)
	Off campus	63 (0.8)	61 (1.5)	63 (0.9)	59 (1.3)	84 (2.2)	60 (1.9)	57 (1.1)
	Both on campus and off campus	11 (0.5)	10 (1.0)	12 (0.6)	12 (0.9)	5 (1.5)	10 (1.2)	14 (0.8)
Time spent at work	Hours per week [mean]	25.74 (0.21)	27.90 (0.46)	24.79 (0.22)	24.67 (0.36)	30.18 (0.66)	23.55 (0.50)	25.53 (0.28)
Pay	Per hour rate [mean]	8.28 (0.06)	9.06 (0.15)	7.93 (0.05)	7.89 (0.07)	7.80 (0.22)	9.17 (0.12)	8.62 (0.09)

Currently work for pay. As we see in Exhibit 17, roughly 52 percent of enrolled Scholars, including 50 and 53 percent of males and females, respectively, worked for pay while in college. With regard to the four minority groups, Asian/Pacific Islanders (38 percent) were less likely than African Americans (56 percent), American Indians/Alaska Natives (56 percent), and Hispanic Americans (52 percent) to work for pay. African Americans were also more likely to work for pay than were Hispanic Americans.

Working on or off campus. Twenty-six percent of the Scholars who worked held on-campus jobs. Many more of the working Scholars (63 percent) worked off campus; 11 percent worked both on and off campus. Male Scholars who were employed were more likely than the employed female Scholars to be working on campus (29 percent vs. 24 percent, respectively).

The patterns of employment among working Scholars varied by minority group. Working American Indians/Alaska Natives (11 percent) were less likely than any of the minority groups to work on campus and, at 84 percent, were more likely than any other group to be working off campus. African American, Asian/Pacific Islander, and Hispanic American Scholars who were working did not differ significantly from each other in their likelihood of working on campus or in their likelihood of working off campus.

Number of hours and rate per hour. Exhibit 17 reveals that employed Scholars worked about 25.74 hours per week, on average. At an average of about 30.18 hours per week, the time spent at work by American Indians/Alaska Natives exceeded that of the other minority groups. Asian/Pacific Islander Scholars, who worked the lowest number of hours per week on average (23.55 hours), worked significantly fewer hours than both American Indian/Alaska Native and Hispanic American Scholars (25.53 hours). Male Scholars who were employed worked an average of 27.90 hours per week, significantly more than working female Scholars, who worked an average of 24.79 hours per week.

The Scholars earned an average of $8.28 per hour, with hourly wages varying by gender and by minority group. Males earned more per hour than females, $9.06 versus $7.93. With an average rate of $9.17 per hour, Asian/Pacific Islanders received more per hour than the other three minority groups. Further, Hispanic American Scholars received more per hour ($8.62) than both African American ($7.89) and American Indian/Alaska Native Scholars ($7.80).

Reasons for working. The Scholars who were working cited several main reasons for doing so (see Exhibit 18). About a third (34 percent) indicated that they were working to meet living expenses. Many Scholars also said they were working to make money (29 percent). Other reasons cited for working were for the learning experience, as part of a work-study program/to pay for school, and to save money.

Exhibit 18. Reasons for Working for the Cohort V Population of Scholars as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Living expenses	Yes	34 (0.8)	30 (1.5)	36 (0.9)	33 (1.3)	36 (2.7)	24 (1.7)	38 (1.1)
	No	66 (0.8)	70 (1.5)	64 (0.9)	67 (1.3)	64 (2.7)	76 (1.7)	62 (1.1)
Learn/ experience	Yes	16 (0.6)	18 (1.2)	15 (0.6)	14 (0.9)	14 (2.1)	29 (1.8)	15 (0.8)
	No	84 (0.6)	82 (1.2)	85 (0.6)	86 (0.9)	86 (2.1)	71 (1.8)	85 (0.8)
Support family	Yes	8 (0.4)	9 (0.9)	7 (0.5)	4 (0.5)	7 (1.4)	12 (1.3)	10 (0.6)
	No	92 (0.4)	91 (0.9)	93 (0.5)	96 (0.5)	93 (1.4)	88 (1.3)	90 (0.6)
Pay back loans/debt	Yes	0 (0.1)	1 (0.2)	0 (0.1)	0 (0.0)	1 (0.5)	0 (0.0)	1 (0.2)
	No	100 (0.1)	99 (0.2)	100 (0.1)	100 (0.0)	99 (0.5)	100 (0.0)	99 (0.2)
Work study/pay for school	Yes	14 (0.5)	11 (0.8)	15 (0.6)	12 (0.9)	8 (1.2)	14 (1.4)	18 (0.8)
	No	86 (0.5)	89 (0.8)	85 (0.6)	88 (0.9)	92 (1.2)	86 (1.4)	82 (0.8)
Social activities	Yes	1 (0.2)	1 (0.2)	2 (0.2)	1 (0.3)	1 (0.5)	0 (0.0)	2 (0.3)
	No	99 (0.2)	99 (0.2)	98 (0.2)	99 (0.3)	99 (0.5)	100 (0.0)	98 (0.3)
Career path/internship	Yes	5 (0.3)	6 (0.6)	4 (0.3)	4 (0.5)	0 (0.0)	12 (1.3)	6 (0.5)
	No	95 (0.3)	94 (0.6)	96 (0.3)	96 (0.5)	100 (0.0)	88 (1.3)	94 (0.5)
Academic requirements	Yes	1 (0.1)	1 (0.2)	1 (0.2)	1 (0.3)	1 (0.5)	0 (0.0)	0 (0.0)
	No	99 (0.1)	99 (0.2)	99 (0.2)	99 (0.3)	99 (0.5)	100 (0.0)	100 (0.0)
Scholarship requirement	Yes	0 (0.1)	1 (0.2)	0 (0.0)	1 (0.2)	0 (0.0)	0 (0.0)	0 (0.0)
	No	100 (0.1)	99 (0.2)	100 (0.0)	99 (0.2)	100 (0.0)	100 (0.0)	100 (0.0)
Save money	Yes	11 (0.5)	10 (0.8)	11 (0.6)	10 (0.8)	8 (1.2)	10 (1.2)	13 (0.7)
	No	89 (0.5)	90 (0.8)	89 (0.6)	90 (0.8)	92 (1.2)	90 (1.2)	87 (0.7)
Make money	Yes	29 (0.7)	30 (1.4)	28 (0.8)	30 (1.2)	28 (2.5)	22 (1.7)	28 (1.0)
	No	71 (0.7)	70 (1.4)	72 (0.8)	70 (1.2)	72 (2.5)	78 (1.7)	72 (1.0)
Other	Yes	10 (0.5)	12 (1.2)	9 (0.5)	11 (0.8)	14 (2.1)	10 (1.2)	7 (0.6)
	No	90 (0.5)	88 (1.2)	91 (0.5)	89 (0.8)	86 (2.1)	90 (1.2)	93 (0.6)

Employed male and female Scholars showed some differences in the reasons they were working. Female Scholars were more likely than males to indicate that they were working to pay living expenses (36 percent vs. 30 percent, respectively) or as part of a work-study program/to pay for school (15 percent vs. 11 percent, respectively). Female Scholars were less likely to be working to learn/for the experience (15 percent vs. 18 percent) or as part of their career path or for an internship (4 percent vs. 6 percent, respectively).

Exhibit 18 reveals some differences between the minority groups in the reasons for working. Asian/Pacific Islanders were the least likely of the minority groups to be working to pay living expenses. At the same time, Asian/Pacific Islanders were more likely than the other minority groups to be working for the learning experience. Asian/Pacific Islanders were also more likely to be working as part of their career path or for an internship compared to African American and Hispanic American Scholars; Hispanic Americans were more likely to be working as part of their career path compared to African Americans. No American Indians/Alaska Natives indicated career path as a reason for working. Hispanic Americans were more likely than African Americans and American Indians/Alaska Natives to be working to save money. Asian/Pacific Islanders were less likely than African Americans and Hispanic Americans to be working to make money.

4.1.2 Sources of Financial Support

As part of the baseline survey, Scholars who were currently enrolled in school were asked a number of questions regarding the type and amount of financial support they were receiving from loans, scholarships, or grants (including the Gates scholarship), and from their parents. In addition, they were asked whether they were able to afford things other students do. Their answers to these questions are displayed in Exhibit 19.

Exhibit 19. Sources of Financial Support for the Cohort V Scholars for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Any loans this academic year	Yes	18 (0.4)	17 (0.8)	18 (0.5)	19 (0.8)	10 (1.2)	18 (1.0)	20 (0.6)
	No	82 (0.4)	83 (0.8)	82 (0.5)	81 (0.8)	90 (1.2)	82 (1.0)	80 (0.6)
Any scholarships or grants this academic year	Yes	98 (0.2)	98 (0.3)	98 (0.2)	97 (0.4)	96 (0.6)	100 (0.0)	99 (0.2)
	No	2 (0.2)	2 (0.3)	2 (0.2)	3 (0.4)	4 (0.6)	0 (0.0)	1 (0.2)
Parents contribute towards college finances	Yes	36 (0.5)	37 (1.1)	35 (0.6)	32 (0.9)	30 (2.0)	45 (1.2)	38 (0.8)
	No	64 (0.5)	63 (1.1)	65 (0.6)	68 (0.9)	70 (2.0)	55 (1.2)	62 (0.8)
Afford things other students do	Yes	54 (0.6)	56 (1.1)	53 (0.7)	53 (1.0)	57 (2.1)	61 (1.2)	51 (0.8)
	No	46 (0.6)	44 (1.1)	47 (0.7)	47 (1.0)	43 (2.1)	39 (1.2)	49 (0.8)

Loans. As we see in Exhibit 19, 18 percent of enrolled Scholars took out loans for the current academic year. American Indians/Alaska Natives were less likely than the Scholars from other minority groups to have taken out a loan for the current academic year. There were no significant differences among the other groups of Scholars in their likelihood of taking a loan.

Scholarships and grants. Ninety-eight percent of enrolled Scholars said that they were the recipients of a scholarship or grant (including the Gates award) for the current academic year. There were no gender differences in reports of receiving scholarships and grants, although there were some differences by minority group. All Asian/Pacific Islander Scholars reported receiving a scholarship or grant. Although almost all Scholars in the other minority groups also indicated that they received a scholarship or grant, the shares of African Americans and American Indians/Alaska Natives reporting this were lower than for Asian/Pacific Islanders and Hispanic Americans.

Parental contribution. Exhibit 19 shows that just over one-third of enrolled Scholars reported that their parents contributed to their college finances. In terms of the four minority groups, at 45 percent, Asian/Pacific Islanders were the most likely to report receiving funds from their parents, followed by Hispanic Americans (38 percent). At 32 and 30 percent, respectively, African Americans and American Indians/Alaska Natives were the least likely of the minority groups to say that their parents helped them out with their college expenses.

Afford what other students do. When asked whether they could afford things that other students do, more than half of the enrolled Scholars (54 percent) answered "yes," with males more likely to answer affirmatively than females (56 percent vs. 53 percent, respectively). Asian/Pacific Islanders (61 percent) were more likely than both African Americans (53 percent) and Hispanic Americans (51 percent) to say that they could afford things that other students do. However, they did not differ significantly from American Indians/Alaska Natives (57 percent). Although American Indians/Alaska Natives were not significantly different from African Americans, they were more likely than Hispanic Americans to say that they could afford what other students do.

Reasons for taking out loans. The 18 percent of Scholars who took out loans were asked the reasons why they did so. Exhibit 20 shows that the reasons Scholars cited most frequently for taking out loans were to pay for books and other educational supplies (59 percent), to pay tuition (57 percent), and to pay for room and board (51 percent). Males were more likely than females to borrow money to pay for books and supplies (66 percent vs. 56 percent). Females were more likely than males to borrow money to pay tuition (59 percent vs. 51 percent). Among the 30 percent of Scholars who indicated that they took out loans to help their family, there were significantly more females (33 percent) than males (23 percent). In addition, among the 14 percent of Scholars who took out loans for extracurricular activities, there were more females (16 percent) than males (9 percent).

Exhibit 20. Reasons for Taking out Loans for the Cohort V Population of Scholars as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Books/other educational supplies	Yes	59 (1.3)	66 (2.4)	56 (1.5)	53 (2.3)	74 (6.0)	59 (3.0)	62 (1.7)
	No	41 (1.3)	34 (2.4)	44 (1.5)	47 (2.3)	26 (6.0)	41 (3.0)	38 (1.7)
Room and board	Yes	51 (1.3)	48 (2.4)	52 (1.6)	42 (2.3)	32 (5.4)	73 (2.7)	56 (1.7)
	No	49 (1.3)	52 (2.4)	48 (1.6)	58 (2.3)	68 (5.4)	27 (2.7)	44 (1.7)
Tuition	Yes	57 (1.3)	51 (2.4)	59 (1.5)	58 (2.3)	42 (6.3)	73 (2.7)	52 (1.7)
	No	43 (1.3)	49 (2.4)	41 (1.5)	42 (2.3)	58 (6.3)	27 (2.7)	48 (1.7)
Extracurricular activities	Yes	14 (0.8)	9 (1.1)	16 (1.1)	9 (1.4)	0 (0.0)	14 (2.0)	21 (1.4)
	No	86 (0.8)	91 (1.1)	84 (1.1)	91 (1.4)	100 (0.0)	86 (2.0)	79 (1.4)
Help family	Yes	30 (1.2)	23 (2.3)	33 (1.4)	36 (2.3)	10 (5.5)	41 (2.9)	25 (1.5)
	No	70 (1.2)	77 (2.3)	67 (1.4)	64 (2.3)	90 (5.5)	59 (2.9)	75 (1.5)
Loan reason: other	Yes	21 (1.1)	17 (1.6)	22 (1.3)	24 (2.0)	16 (4.0)	9 (1.7)	24 (1.5)
	No	79 (1.1)	83 (1.6)	78 (1.3)	76 (2.0)	84 (4.0)	91 (1.7)	76 (1.5)

Among the minority groups, African Americans were less likely than American Indians/Alaska Natives and Hispanic Americans to borrow money for books and supplies. Asian/Pacific Islanders were more likely than the other minority groups to borrow money to pay for room and board. Hispanic Americans were also more likely than African Americans and American Indians/Alaska Natives to borrow money for room and board. Asian/Pacific Islanders were more likely than the other minority groups to take out loans to pay tuition. Hispanic Americans were more likely than African Americans and Asian/Pacific Islanders to take out loans to pay for extracurricular activities. Since no American Indians/Alaska Natives reported taking out loans to pay for extracurricular activities and the s.e. equals zero, statistical comparisons could not be made between them and other minority groups. African Americans and Asian/Pacific Islanders were not significantly different in their rate of borrowing money to help their family; both were more likely to do so than American Indians/Alaska Natives and Hispanic Americans.

4.2 *Early Outcomes and Experiences of the Non-Recipients*

4.2.1 Work for Pay

The survey asked non-recipients who were currently enrolled in college a series of questions regarding their work activities. Exhibits 21 and 22 display the answers they gave to these questions.

Exhibit 21. Work Activities of the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) or Mean (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent or Mean (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Currently work for pay	Yes	67 (1.3)	62 (2.7)	69 (1.5)	69 (2.4)	63 (4.2)	64 (2.7)	69 (2.1)
	No	33 (1.3)	38 (2.7)	31 (1.5)	31 (2.4)	37 (4.2)	36 (2.7)	31 (2.1)
Work on or off campus	On campus	26 (1.5)	25 (3.1)	27 (1.7)	32 (2.9)	19 (4.4)	28 (3.2)	23 (2.3)
	Off campus	60 (1.7)	64 (3.4)	59 (1.9)	54 (3.1)	71 (5.1)	59 (3.5)	64 (2.6)
	Both on campus and off campus	13 (1.1)	10 (2.2)	14 (1.3)	14 (2.1)	10 (3.5)	13 (2.4)	13 (1.8)
Time spent at work	Hours per week [mean]	26.75 (0.46)	28.49 (1.02)	26.13 (0.500)	27.21 (0.83)	30.70 (1.11)	23.15 (0.95)	27.16 (0.76)
Pay	Per hour rate [mean]	8.08 (0.12)	8.27 (0.21)	8.01 (0.14)	7.87 (0.17)	7.56 (0.26)	8.83 (0.47)	8.01 (0.12)

Currently work for pay. Exhibit 21 shows that 67 percent of enrolled non-recipients worked for pay during the school year, a rate that is considerably higher than the rate for Scholars. Although the employment rates of the non-recipients did not differ significantly by minority group, female non-recipients were more likely to work for pay than males (69 percent vs. 62 percent, respectively). Comparing across the Scholar and non-recipient populations, we found that significantly more non-recipients were working for pay than Scholars for every minority group except American Indians/Alaska Natives (30.18 hours).

Working on or off campus. Of the non-recipients who were working, 26 percent held jobs on campus, 60 percent worked off campus, and 13 percent worked both on and off campus. Overall, Scholars and non-recipients did not differ in the rates at which they worked on versus off campus, or both; there were no significant differences by minority group.

There were no statistically significant differences by gender in the shares of non-recipients working on, off, or both on and off campus. In terms of the minority groups, relatively more American Indians/Alaska Natives than African Americans held off-campus jobs. This is consistent with the pattern found for Scholars, in which American Indians/Alaska Natives were more likely than Scholars in the other minority groups to work off campus.

Number of hours and rate per hour. Exhibit 21 shows that, on average, employed non-recipients worked about 26.75 hours per week. Male non-recipients worked more hours per week than female non-recipients.

There were also some differences in hours worked by minority group. American Indian/Alaska Native non-recipients spent more time on average working compared to other minority groups; however, the difference was statistically significant only in comparison to Asian/Pacific Islanders. Asian/Pacific Islanders spent less time working than the other non-recipient groups. At the level of the populations, employed Scholars and non-recipients did not differ significantly in hours worked per week. However, at the level of the minority group, African American non-recipients worked more hours per week than their Scholar counterparts.

On average, non-recipients earned $8.08 per hour. The Scholars and non-recipients did not differ significantly in rate of pay per hour nor were there differences among the non-recipients by minority group. When Scholars and non-recipients were compared by minority group, we found that Hispanic American non-recipients earned less than their Scholar counterparts did; however, there were no significant differences for the other minority groups.

Reasons for working. As Exhibit 22 shows, non-recipients who were working for pay did so primarily as part of a work-study program/to pay for school (41 percent), to pay for living expenses (35 percent), and to make money (26 percent). Other reasons for working for pay were endorsed by 10 percent or fewer of the non-recipients. The only gender difference in the reasons non-recipients were working was that females were more likely than males to be working to learn or for the experience (9 percent vs. 5 percent, respectively).

Overall, there were few differences by racial minority group in reasons the non-recipients were working. Asian/Pacific Islanders were more likely than Hispanic Americans to be working for the learning experience. Hispanic Americans were more likely than African Americans to be working as part of a work-study program/to pay for school.

Compared to the Scholars, non-recipients were more likely to indicate that they were working to pay back loans or debt or were engaged in a work-study program/working to pay for school. Scholars were more likely than non-recipients to be working for the learning experience, to support their family, or as part of their career path or internship. These differences between the Scholars and non-recipients in reasons for working apply across gender and minority groups with few exceptions. The exceptions are that American Indian/Alaska Native Scholars and non-recipients show no significant differences in the rates they were working for the experience, to support family, or to pay back loans or debt. In addition, African American Scholars and non-recipients are similar in the rates they worked to support family. Although 3 percent of African American non-recipients worked to pay back loans or debt, no statistical comparison could be made because none of their Scholar counterparts reported working for this reason. Further, although 3 percent of American Indian/Alaska Native non-recipients reported working as part of their career path or for an internship, none of their Scholar counterparts reported this as a reason, preventing a statistical comparison.

Exhibit 22. Reasons for Working for the Cohort V Population of Non-Recipients as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Living expenses	Yes	35 (1.6)	36 (3.5)	34 (1.8)	30 (2.9)	40 (5.5)	30 (3.3)	40 (2.7)
	No	65 (1.6)	64 (3.5)	66 (1.8)	70 (2.9)	60 (5.5)	70 (3.3)	60 (2.7)
Learn/experience	Yes	8 (0.9)	5 (1.5)	9 (1.1)	8 (1.6)	10 (3.2)	15 (2.6)	4 (1.1)
	No	92 (0.9)	95 (1.5)	91 (1.1)	92 (1.6)	90 (3.2)	85 (2.6)	96 (1.1)
Support family	Yes	4 (0.7)	3 (1.3)	4 (0.8)	3 (1.1)	2 (1.6)	5 (1.6)	4 (1.1)
	No	96 (0.7)	97 (1.3)	96 (0.8)	97 (1.1)	98 (1.6)	95 (1.6)	96 (1.1)
Pay back loans/debt	Yes	4 (0.7)	4 (1.5)	4 (0.7)	3 (1.1)	2 (1.6)	7 (1.8)	3 (1.0)
	No	96 (0.7)	96 (1.5)	96 (0.7)	97 (1.1)	98 (1.6)	93 (1.8)	97 (1.0)
Work study/pay for school	Yes	41 (1.7)	46 (3.5)	40 (1.9)	36 (2.9)	39 (5.5)	42 (3.6)	47 (2.7)
	No	59 (1.7)	54 (3.5)	60 (1.9)	64 (2.9)	61 (5.5)	58 (3.6)	53 (2.7)
Social activities	Yes	1 (0.3)	0 (0.0)	1 (0.4)	1 (0.6)	0 (0.0)	1 (0.6)	1 (0.6)
	No	99 (0.3)	100 (0.0)	99 (0.4)	99 (0.6)	100 (0.0)	99 (0.6)	99 (0.6)
Career path/internship	Yes	2 (0.4)	1 (0.7)	2 (0.5)	1 (0.6)	3 (2.0)	3 (1.2)	2 (0.7)
	No	98 (0.4)	99 (0.7)	98 (0.5)	99 (0.6)	97 (2.0)	97 (1.2)	98 (0.7)
Academic requirements	No	100 (0.0)	100 (0.0)	100 (0.0)	100 (0.0)	100 (0.0)	100 (0.0)	100 (0.0)
Scholarship requirement	Yes	0 (0.2)	0 (0.0)	0 (0.2)	0 (0.4)	0 (0.0)	1 (0.6)	0 (0.0)
	No	100 (0.2)	100 (0.0)	100 (0.2)	100 (0.4)	100 (0.0)	99 (0.6)	100 (0.0)
Save money	Yes	10 (1.0)	11 (2.2)	10 (1.2)	14 (2.1)	16 (4.1)	7 (1.9)	8 (1.5)
	No	90 (1.0)	89 (2.2)	90 (1.2)	86 (2.1)	84 (4.1)	93 (1.9)	92 (1.5)
Make money	Yes	26 (1.5)	26 (3.2)	26 (1.7)	29 (2.8)	25 (4.9)	23 (3.0)	25 (2.4)
	No	74 (1.5)	74 (3.2)	74 (1.7)	71 (2.8)	75 (4.9)	77 (3.0)	75 (2.4)
Other	Yes	8 (0.9)	9 (2.0)	8 (1.0)	9 (1.8)	7 (2.7)	13 (2.4)	5 (1.2)
	No	92 (0.9)	91 (2.0)	92 (1.0)	91 (1.8)	93 (2.7)	87 (2.4)	95 (1.2)

4.2.2 Sources of Financial Support

Non-recipients who were enrolled in college or university at the time of the baseline survey were also asked a number of questions regarding the financial support they were currently receiving from loans, scholarships, grants, and their parents, and about whether they could afford the things that other students do. Their answers to these questions are displayed in Exhibit 23.

Loans. Exhibit 23 shows that 59 percent of enrolled non-recipients received support from loans during the most recent academic year, a much higher share than that of the Scholars at 18 percent.

Exhibit 23. Sources of Financial Support for the Cohort V Non-Recipients for the Population as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Any loans this academic year	Yes	59 (1.4)	61 (2.8)	58 (1.6)	73 (2.3)	42 (4.5)	59 (2.9)	51 (2.3)
	No	41 (1.4)	39 (2.8)	42 (1.6)	27 (2.3)	58 (4.5)	41 (2.9)	49 (2.3)
Any scholarships or grants this academic year	Yes	89 (0.9)	91 (1.7)	88 (1.1)	90 (1.6)	84 (3.4)	90 (1.7)	88 (1.5)
	No	11 (0.9)	9 (1.7)	12 (1.1)	10 (1.6)	16 (3.4)	10 (1.7)	12 (1.5)
Parents contribute towards college finances	Yes	55 (1.4)	59 (2.8)	53 (1.6)	56 (2.6)	43 (4.4)	57 (2.8)	55 (2.2)
	No	45 (1.4)	41 (2.8)	47 (1.6)	44 (2.6)	57 (4.4)	43 (2.8)	45 (2.2)
Afford things other students do	Yes	48 (1.4)	50 (2.8)	47 (1.6)	44 (2.6)	58 (4.4)	53 (2.9)	46 (2.3)
	No	52 (1.4)	50 (2.8)	53 (1.6)	56 (2.6)	42 (4.4)	47 (2.9)	54 (2.3)

There were no statistically significant differences by gender in the non-recipients' tendency to take out loans for the current year.

In terms of the four minority groups, African American non-recipients, with close to three-quarters securing loans, borrowed funds at a higher rate than their non-recipient minority counterparts. The next highest rate of borrowing was by the Asian/Pacific Islander non-recipients whose rate of borrowing was significantly higher than the lowest group, American Indians/Alaska Natives. Among both Scholars and non-recipients, fewer American Indian/Alaska Native students took out loans for the current academic year.

Across all minority groups, Scholars were less likely than their non-recipient counterparts to take out loans in the current academic year.

Scholarships and grants. Eighty-nine percent of enrolled non-recipients reported that they were the recipient of a scholarship or grant for the current academic year. As a whole, non-recipients were less likely than were their Scholar counterparts (98 percent) to receive awards.

The relative shares of non-recipients receiving awards for the current academic year did not differ significantly by gender or by minority group.

Compared to the Scholars, non-recipients in every minority group except Asian/Pacific Islanders were less likely than Scholars to receive scholarships or grants for the current academic year. Although fewer Asian/Pacific Islander non-recipients received scholarships or grants (90 percent) compared to their Scholar counterparts (100 percent), lack of variation for this statistic (s.e. equals zero) in the Asian/Pacific Islander Scholar sample prevented statistical comparison.

Parental contribution. A little more than half of enrolled non-recipients reported that their parents contributed to their college finances, a proportion that is well above that of the Scholars (55 percent vs. 36 percent, respectively). Although there were no significant gender differences in rates that parents contributed towards college, the difference in the rates at which Asian/Pacific Islanders (57 percent) and American Indians/Alaska Natives (43 percent) received parental contributions were significantly different. No other differences in parental contribution by minority group were significant. Compared to the Scholars, non-recipients in every minority group were more likely than the Scholars to receive parental contributions for college.

Afford what other students do. When asked whether they could afford things that other students do, just under half of enrolled non-recipients responded that they could. Although there were no gender differences, the difference between the group reporting the highest and lowest rates of being able to afford what other students do was significant (American Indians/Alaska Natives at 58 percent vs. African Americans at 44 percent).

Compared to the Scholars, the non-recipients, as a whole, were less likely to say that they could afford what other students do. This difference is largely due to the African American and Asian/Pacific Islander Scholars, who were more likely than were their non-recipient counterparts to report that they could afford what other students do.

Reasons for taking out loans. As Exhibit 24 shows, the non-recipients who took out loans did so primarily to pay for tuition (79 percent), room and board (65 percent), and books and other educational supplies (61 percent). When they took out loans, the non-recipients were more likely to borrow money to pay for tuition and for room and board than were the Scholars. Scholars were more likely to borrow money to pay for "other" expenses, with 21 percent of Scholars citing other reasons for taking loans, compared to 5 percent of non-recipients.

Thirty-five percent of Asian/Pacific Islander non-recipients who took out loans borrowed money to help family, a rate that was significantly higher than that for African Americans (21 percent), who borrowed least often to help family. No other differences by minority group were significant. The only significant gender difference was that male non-recipients cited paying for extracurricular activities (16 percent) as a reason for taking out loans more often than did female non-recipients (10 percent).

Some differences by minority group in the reasons why Scholars and non-recipients took out loans are apparent. African American non-recipients were more likely than the Scholars to take out loans for books/educational supplies, room and board, and tuition; however, they were less likely to take out loans to help family. American Indian/Alaska Native non-recipients were more likely than were American Indian/Alaska Native Scholars to take out loans for room and board and for tuition. Compared to their Scholar counterparts, Hispanic American non-recipients were more likely to take out loans to pay for tuition but less likely to do so to pay for extracurricular activities. For Asian/Pacific Islanders, Scholars and non-recipients did not differ significantly in the reasons they were taking out loans.

Exhibit 24. Reasons for Taking out Loans for the Cohort V Population of Non-Recipients as a Whole and by Gender and Minority Group [Percent (s.e.) in each Category]

		Population	Gender		Minority Group			
		Percent (s.e.)	Male	Female	African Americans	American Indians	Asian/Pacific Islanders	Hispanic Americans
Books/other educational supplies	Yes	61 (1.8)	59 (3.6)	61 (2.1)	63 (3.0)	60 (6.8)	61 (3.8)	58 (3.2)
	No	39 (1.8)	41 (3.6)	39 (2.1)	37 (3.0)	40 (6.8)	39 (3.8)	42 (3.2)
Room and board	Yes	65 (1.8)	68 (3.5)	63 (2.1)	69 (2.9)	54 (6.9)	62 (3.7)	63 (3.1)
	No	35 (1.8)	32 (3.5)	37 (2.1)	31 (2.9)	46 (6.9)	38 (3.7)	37 (3.1)
Tuition	Yes	79 (1.6)	74 (3.3)	81 (1.7)	77 (2.7)	67 (6.5)	81 (3.0)	81 (2.6)
	No	21 (1.6)	26 (3.3)	19 (1.7)	23 (2.7)	33 (6.5)	19 (3.0)	19 (2.6)
Extracurricular activities	Yes	12 (1.2)	16 (2.6)	10 (1.3)	10 (1.8)	8 (3.6)	18 (2.9)	12 (2.1)
	No	88 (1.2)	84 (2.6)	90 (1.3)	90 (1.8)	92 (3.6)	82 (2.9)	88 (2.1)
Help family	Yes	27 (1.6)	29 (3.3)	26 (1.9)	21 (2.5)	24 (5.9)	35 (3.6)	31 (3.0)
	No	73 (1.6)	71 (3.3)	74 (1.9)	79 (2.5)	76 (5.9)	65 (3.6)	69 (3.0)
Loan reason: other	Yes	5 (0.8)	5 (1.6)	6 (1.0)	4 (1.2)	13 (4.7)	4 (1.5)	6 (1.6)
	No	95 (0.8)	95 (1.6)	94 (1.0)	96 (1.2)	87 (4.7)	96 (1.5)	94 (1.6)

APPENDICES

Appendix A:

Sample Design for the Non-Recipients

SAMPLE DESIGN FOR THE NON-RECIPIENTS

In Year 4 of the longitudinal study, the population of Cohort V Scholars and a sample of Cohort V non-recipients were asked to complete the baseline survey, marking their first year of participation in the study.

The population of Cohort V Scholars is defined as all applicants who applied to the program for aid as entering freshmen, were offered scholarships, and accepted them for their freshman year in college, in this case the 2004-2005 academic year and the fifth award year of the scholarship program.

The population of non-recipients, on the other hand, is defined as all applicants who applied to the program for aid as entering freshmen for the same award year (2004-2005) as the Scholars, made it through the reader selection process, but did not receive scholarships for one or more reasons.

In the spring of 2005, the United Negro College Fund (UNCF) supplied NORC with a preliminary data file that included records for all Cohort V nominees who made it through the reader selection process. After some additional processing by UNCF, NORC received the final file from UNCF on June 2, 2005.

The final file contained a record for each of 4,464 nominees, 1,000 Scholars, and 3,464 non-recipients and included all information currently available in machine-readable form in the UNCF database. In addition to contact information and minority group, it contained selection status (Select/Non-Select), scholarship status (recipient/non-recipient), and gender for all nominees.

Exhibit A.1 displays a break out of the non-recipient population by minority group, gender, and selection status. Although the non-recipient population originally included 3,464 nominees, one sample member, a female African American Non-Select, was reported to be deceased at the time of the baseline survey. The population sizes shown in Exhibit A.1 have been adjusted accordingly.

Exhibit A.1. Population Counts for the Cohort V Non-Recipients Who Made it Through the Reader Selection Process

Minority Group	Gender	Selection Status		Total
		Selects	Non-Selects	
African American	Males	65	188	253
	Females	133	707(1)*	840(1)*
American Indian	Males	37	66	103
	Females	87	164	251
Asian/Pacific Islander	Males	28	163	191
	Females	57	437	494
Hispanic American	Males	52	378	430
	Females	87	814	901
Total		546	2917(1)*	3,463(1)*

*The number in parentheses indicates that one population member was known to be deceased at the time of the baseline survey and has been removed from the counts in the exhibit.

To draw a sample of Cohort V non-recipients for participation in the survey, NORC relied on a proportionate stratified sampling procedure. The desired sample size for each cell of the minority group by gender by selection status design was computed by dividing the desired sample size of 1,333 non-recipients by the size of the non-recipient population (3,464) and multiplying by the subpopulation size within the cell. The sample members within each cell of the design were drawn using a systematic sampling procedure in which the records were sorted by state and zip code of permanent residence before the sample was drawn.

Exhibit A.2 shows the number of cases selected to participate in the survey within each cell of the 4 by 2 by 2 sample design. Again, the counts in the exhibit are adjusted for the deceased member in the sample.

Exhibit A.2. Sample Counts for the Cohort V Non-Recipients Who Made it Through the Reader Selection Process

Minority Group	Gender	Selection Status		Total
		Selects	Non-Selects	
African American	Males	25	72	97
	Females	51	271(1)*	322(1)*
American Indian	Males	14	25	39
	Females	34	63	97
Asian/Pacific Islander	Males	11	63	74
	Females	22	168	190
Hispanic American	Males	20	146	166
	Females	34	313	347
Total		211	1121(1)*	1332(1)*

*The number in parentheses indicates that one sample member was known to be deceased at the time of the baseline survey and has been removed from the counts in the exhibit.

APPENDIX B:

SURVEY METHODOLOGY

SURVEY METHODOLOGY

B.1 Instrument Design and Development

NORC developed the baseline questionnaire in collaboration with the Foundation and members of the RAC. The instrument included the same items administered to Cohorts I, II, and III to facilitate cross-cohort comparisons. It also contained some newly developed items. The questionnaire was designed to collect information in several broad areas of interest:

- educational background and high school experiences,
- academic and civic engagement (leadership),
- transition to college,
- academic self-esteem and locus of control,
- early outcomes and experiences while in college,
- attitudes and beliefs,
- demographic background, and
- experiences with the GMS program.

To the extent possible, NORC drew the items for the instrument from the National Education Longitudinal Survey: 88 Second Follow-Up, Baccalaureate and Beyond Second Follow-Up, High School and Beyond, College and Beyond, General Social Survey, Survey of Earned Doctorates 2002, and Survey of Doctorate Recipients 2001. For areas not covered in those surveys, NORC developed new questions and tested them by phone in cognitive interviews with a sample of male and female Scholars and non-recipients from each of the four minority groups.

Designed for self-administration on the web, the 30-minute instrument was divided into eight distinct sections presented to the Scholars and non-recipients in the following order:

- College Enrollment
- Gates (Experience with GMS Program)
- Work and College Finances
- Academic and Community Engagement
- College Experience
- Your Attitudes (Self-Assessment)
- Future Plans
- Background Information

Once the questions were entered into the SCYWEB software system for administration on the web, NORC conducted extensive testing on the instrument to ensure that the skip patterns were working as intended and that the instrument could be completed within 30 minutes.

B.2 Data Collection Procedures

Baseline data were collected for Cohort V from June 6, 2005 through October 3, 2005. Three principal goals guided this effort:

- To obtain the highest possible degree of participation from Cohort V recipients and non-recipients;
- To target additional efforts to contact hard-to-reach respondents; and
- To obtain high-quality data from each respondent.

Each of these goals contributed to the success of the fourth year of the project.

In order to boost response rates, NORC adopted a multi-modal approach to prompting that used mail, telephone, and e-mail contact strategies with high rates of effectiveness.

NORC began the data collection period by mailing an introductory letter to Scholars and non-recipients, notifying them that they had been chosen for the study and inviting them to participate. The Scholar letter described the study and its purpose, and provided a unique PIN and password to use in accessing the survey online. In addition to including the items in the Scholar letter, the letter for the non-recipient population offered a monetary incentive of $25 upon completion of the web-based survey.

NORC followed up the introductory letter mailings with a series of prompt letters and e-mails reminding sample members to complete the survey. When NORC received confirmation that sample members no longer resided at a particular location, we attempted to locate the case. Table B.1 provides a complete list of prompting techniques as well as a timeline of when they were implemented.

Exhibit B.1. Prompting Efforts and Timeline for the Cohort V Baseline Survey

Prompting Task	Date
Cohort V Advance Letter	June 15, 2005
Cohort V Recipient Batch E-mail	June 20, 2005
Cohort V Non-Recipient Batch E-mail	June 21, 2005
Cohort V Postcard Mailing	June 27, 2005
Cohort V Follow-Up Postcard Letter	June 30, 2005
Gates Foundation Letter (recipients only)	July 15, 2005
Cohort V Phone Prompting	July 18, 2005
Newly Located Advance Letter	July 20, 2005
Prompt Letter	August 5, 2005
2nd Batch E-mail Prompt	August 23, 2005-August 24, 2005

In an effort to further boost response rates for the American Indian populations, NORC contracted with Paradox, a consulting firm specializing in American Indian issues, to assist in prompting the American Indian population for Cohort V. (For a more detailed description of this effort, see Bronwyn Nichols Lodato, Michele Zimowski, Rashna Ghadialy, and Lekha Venkataraman, *Gates Millennium Scholars Tracking and Longitudinal Study Year 4 Final Report* [Chicago: National Opinion Research Center, 2007], http://www.norc.org/gatesscholars.)

In addition to the batch e-mails, prompting letters, and phone prompting employed throughout data collection, NORC also relied on a tailored e-mail strategy. This strategy utilized customized e-mail text to reach two specific target populations. The first of these populations included respondents from Cohort V who had started the survey but had not yet completed it (suspended cases). The second group to receive these tailored e-mails included two randomly selected replicates of Scholars and two randomly selected replicates of non-recipients.[21] These selected Cohort V replicates received all the prompt letters as well as e-mail prompts, but were not included in phone prompting, in an effort to investigate the effectiveness of e-mail prompting versus phone prompting. (For details of the tailored e-mail prompting effort, see Bronwyn Nichols Lodato, Michele Zimowski, Rashna Ghadialy, and Lekha Venkataraman, *Gates Millennium Scholars Tracking and Longitudinal Study Year 4 Final Report* [Chicago: National Opinion Research Center, 2007], http://www.norc.org/gatesscholars.)

[21] Both the Scholar population and non-recipient sample were randomly divided into ten replicates of roughly equal size in advance of the baseline survey. The purpose of creating these replicates is to enable us to conduct methodological studies on a representative sample of cases. Each Scholar replicate consists of a representative sample of about 100 Scholars. Each non-recipient replicate consists of a representative sample of about 133 non-recipients.

Appendix C:

Response Rates

Response Rates

The overall response rates of Cohort V to the baseline survey are shown in Exhibit C.1 for the population of Scholars and the sample of non-recipients. The exhibit shows that close to 89 percent of Scholars and 73 percent of the non-recipient sample completed the baseline survey. Exhibit C.2 reveals that the rates of responding by minority group ranged from 0.81 to 0.92 in the Scholar population and from 0.69 to 0.81 in the non-recipient sample. Exhibit C.3, on the other hand, shows that, as a whole, females were more likely than their male counterparts to complete the survey, but much more so in the sample of non-recipients than in the population of Scholars. Finally, Exhibit C.4 reveals that non-recipients who were selected to go on to the verification/confirmation phase in the Scholar selection process were a little more likely to respond to the survey than those who were not asked to go on.

Exhibit C.1. Response Rates for the Baseline Survey of Cohort V by Population Type*

Population	Population/Sample Size Baseline	Baseline Number of Completes	Baseline Response Rates
2004 Freshmen Scholars	1000	890	0.89
2004 Freshmen Non-Recipients	1332**	967	0.73
Total	2332**		

*The response rates for the non-recipients are weighted to account for small differences in the selection probabilities of the sample members. No such adjustment was necessary for the Scholars since all Scholars were selected for the survey with certainty.

**The sample member who was deceased at the time of the baseline survey is not included in the counts.

Exhibit C.2. Response Rates for the Baseline Survey of Cohort V by Population Type and Minority Group*

Population	Minority Group	Population/Sample Size Baseline	Baseline Number of Completes	Baseline Response Rates
2004-2005 Freshmen Scholars	African American	350	308	0.88
	American Indian	150	121	0.81
	Asian/Pacific Islander	150	138	0.92
	Hispanic American	350	323	0.92
2004-2005 Freshmen Non-Recipients	African American	419**	287	0.69
	American Indian	136	100	0.74
	Asian/Pacific Islander	264	215	0.81
	Hispanic American	513	365	0.71

*The response rates for the non-recipients are weighted to account for small differences in the selection probabilities of the non-recipient sample members. No such adjustment was necessary for the Scholars since all Scholars were selected for the survey with certainty.

**The sample member who was deceased at the time of the baseline survey is not included in the count.

Exhibit C.3. Response Rates for the Baseline Survey of Cohort V by Population Type and Gender*

Population	Gender	Population/Sample Size Baseline	Baseline Number of Completes	Baseline Response Rates
2004 Freshmen Scholars	Males	313	275	0.88
	Females	687	615	0.90
2004 Freshmen Non-Recipients	Males	376	247	0.66
	Females	956**	720	0.75

*The response rates for the non-recipients are weighted to account for small differences in the selection probabilities of the sample members. No such adjustment was necessary for the Scholars since all Scholars were selected for the survey with certainty.

**The sample member who was deceased at the time of the baseline survey is not included in the count.

Exhibit C.4. Response Rates for the Baseline Survey of Cohort V Non-Recipients by Selection Status*

Population	Selection Status	Sample Size Baseline	Baseline Number of Completes	Baseline Response Rates
2004 Freshmen Non-Recipients	Select	211	162	0.77
	Non-Select	1121**	805	0.72

*The response rates for the non-recipients are weighted to account for small differences in the selection probabilities of the sample members.

**The sample member who was deceased at the time of the baseline survey is not included in the count.

Appendix D:

Survey Weights

SURVEY WEIGHTS

At the end of the field period, NORC generated two separate sets of cross-sectional survey weights for Cohort V, one for the Scholar population, and the other for the non-recipient population. Weights were needed for two main reasons: to compensate for small differences in the selection probabilities of the non-recipient sample members and to compensate for subgroup differences in the participation rates of the Scholar population and the non-recipient sample.

The weighting procedure for the Scholar population consisted of two steps. The procedure for non-recipients consisted of three steps. The first step, applied to both populations, computed a base weight for each member in the sample reflecting his or her probability (chances) of selection into the survey. In the Scholar population, each member had the same selection probability; all Scholars were asked to participate in the survey. In the non-recipient population, the selection probabilities differed slightly across the cells of` the minority group by gender by selection status design since they were based on the rounded number of desired sample members in each cell.

The second step, again applied to both populations, adjusted the base weights for non-response to the baseline survey within each cell of an adjustment design. The adjustment cells for each population were selected on the basis of information available in the UNCF databases. For the Scholar population, cells were defined by minority group and gender. For the non-recipient population, the cells were defined by minority group, gender, and selection status (Select/Non-Select). The non-response-adjusted weight for a case is simply the base weight inflated by the inverse of the response rate for the adjustment cell for that case.

The final step of the weighting procedure, applied only to the non-recipient population, adjusted for the departure from population totals due to the death of a sample member prior to the onset of the survey. In this step, the weights were post-stratified to bring them in line with the population totals in the minority group by gender by Select/Non-Select cells after adjusting the totals for the deceased member.

The procedure is designed to yield a set of weights for completed cases that sums to the number of Scholars or the number of non-recipients in the respective populations. Similarly, the weights within each stratum of the sampling design for the Scholar and non-recipient populations were designed to sum to the subpopulation total within the cell.

APPENDIX E:

STANDARD ERRORS

STANDARD ERRORS

Standard errors indicate the degree of uncertainty associated with population estimates obtained from a sample draw of a population. In the case of the GMS survey, sampling error or variance applies to estimates obtained from the samples of non-recipients, but not to those of the Scholar populations since all members of the Scholar populations were asked to take part in the survey. Nonetheless, there is still uncertainty associated with the estimates for the Scholar populations due to non-response to the survey—not all Scholars completed the survey.

Another factor that affects the calculation of standard errors is the size of the sample relative to the size of the population. When data are collected from a substantial proportion of the population, other things being equal, there is less uncertainty associated with the estimates. In the computation of standard errors, this extra precision is taken into account with a factor known as the finite population correction.[22] When a sample includes less than 5 percent of the population, the finite population correction factor is typically ignored because it has a negligible effect on the size of the standard errors. However, as the sample size approaches the size of the population, the effect of the correction becomes more and more noticeable. As a rule of thumb, Cochran recommends incorporating the correction when the sample accounts for more than 5 or 10 percent of the population (see William Cochran, *Sampling Techniques*, 3rd ed. [New York: Wiley, 1977]). In the case of the GMS Survey, the number of respondents in the non-recipient and Scholar populations well exceeds that threshold.

Standard errors calculated by most conventional statistical routines are based on the assumptions of simple random sampling (SRS) and an infinite population size, assumptions that are not entirely warranted in this study because of the complex sample design and the size of the samples relative to the size of the populations.[23]

The effect of complex sample designs on the standard errors are typically incorporated in reported results in one of two main ways. The first is to use one or a set of general correction factors as adjustments to the standard errors estimated assuming SRS. This has been the standard practice of analysts using data from many of the large national education sample surveys such as High School and Beyond, National Post-Secondary Student Aid Survey, Baccalaureate and Beyond, and National Education Longitudinal Study of 1988. The public documentation of these projects contains general correction factors that analysts can draw upon readily.

The second method relies on specialized software packages to directly calculate the design effect-adjusted standard errors. Packages of this sort use a variety of different statistical algorithms to make the necessary adjustments—the most common of which are the Taylor Series Approximation, Balanced Repeated Replication, and Jackknife Repeated Replication. As a rule, NORC projects rely on the Taylor Series Approximation method as implemented in the SUDAAN statistical software package, a package that also incorporates the finite population correction factor. Calculations of this type, along with the finite population correction factor, were applied to the survey data to generate the standard errors found in this and other GMS Main Findings Reports.

Within SUDAAN, strata were defined to be consistent with the sampling plan for non-recipients and the non-response adjustment cells for Scholars. All calculations were weighted by the adjusted sample weights.

[22] Note that the finite population correction only applies to situations where sampling is done without replacement.

[23] SPSS and standard SAS routines assume an infinite population. SAS, however, offers a series of specialized routines for complex sampling plans (e.g., SURVEYFREQ, SURVEYMEANS, SURVEYREG, and SURVEYLOGISTIC) capable of incorporating a finite population correction. SPSS also offers some specialized routines.

The standard error for an estimated percent or mean can be used to construct a confidence interval around the estimate. A confidence interval is a range of values that includes, with a specified level of confidence, the population value. So, for example, we would have 95 percent confidence that the 95 percent confidence interval includes the population value. Of course, any specific confidence interval may or may not actually include the population value, but the logic is that if the study were replicated again and again, then, on average, 95 percent of the generated 95 percent confidence intervals would include the population value.

When the sample size is 500 or more, a 95 percent confidence interval is calculated as the estimate plus or minus 1.96 times the standard error. As an illustration, take the percent of female Scholars whose father's educational attainment was less than high school (Exhibit 5, Educational Attainment). The estimated percent is 24 and the standard error is 0.5. The 95 percent confidence interval (where x represents the population value) is

$$24 - 1.96 * 0.5 \le x \le 24 + 1.96 * 0.5$$

$$23.02 \le x \le 24.98$$

In this example, we would say that our estimate of the percent of female Scholars whose father's educational attainment was less than high school is 24 percent. In addition, we are 95 percent confident that the population value is between 23.02 percent and 24.98 percent. So the population value may well be 23.5 percent or 24.5 percent, but is very unlikely to be 35 percent.

The standard error of the difference between two sample estimates taken from different groups can be calculated using the standard errors of the two estimates

$$s_{x-y} = \sqrt{s_x^2 + s_y^2},$$

where s_{x-y} is the standard error of the difference of the sample estimates, and s_x and s_y are the standard errors of the estimates, x and y. Estimated means or percentages can be compared this way.

As an illustration (Exhibit 5, Educational Attainment), take the difference between the estimated percent of Hispanic American Scholars whose father's educational attainment was less than high school (44 percent with a standard error of 0.8) and the corresponding estimate for African Americans (12 percent with a standard error of 0.7). The standard error of the difference between the estimated percents is:

$$\sqrt{0.8^2 + 0.7^2} = 1.1$$

The estimated difference between the two groups is 32 (44 - 12). A 95 percent confidence interval is calculated as the estimate plus or minus 1.96 times the standard error. In this case, the confidence interval of the difference in estimated percents would be 32 plus or minus 1.1 times 1.96, or (where $x - y$ represents the population difference):

$$29.8\% \le x - y \le 34.2\%$$

Since the 95 percent confidence interval for the difference does not include 0, we would conclude that these two groups have a statistically significant (at the .05 level) difference in the percent of fathers with less than a high school diploma, and our estimate of that difference is 32 percent. In this way, standard errors can be used to test for group differences in means or percents appearing in the exhibits. The 1.96 multiplier is based on a 95 percent confidence interval and assumes a pooled sample size of 500 or more.

If multiple differences are calculated and used for significance testing—for example the differences between the enrollment rates of Hispanic Americans and Asian/Pacific Islanders, Asian/Pacific Islanders and American Indians/Alaska Natives, and American Indians/Alaska Natives and African Americans—then the multiplier can be increased from 1.96 to reflect the number of comparisons (number of tests) made. This is equivalent to the Bonferroni adjustment used in multiple comparison testing (see Jason C. Hsu, *Multiple Comparisons: Theory and Method* [London: Chapman & Hall, 1996] for a discussion of the Bonferroni adjustment). Appendix F discusses how we applied the adjustment in this report.

It is important to note that the percent of respondents in some of the response categories is 0 or 100 percent. In those cases, the exhibits report the standard error as zero since there is no variation in the sample.[24] Because there is no variation, SUDAAN is unable to estimate standard errors for those percents. Consequently, it is not possible to compare them statistically with other values in the exhibits.

In cases where the *estimated* percent in a particular response category is 0 or 100 percent, the standard errors have non-zero values in the survey exhibits.

Also note that in some cases the percents for a survey question do not sum to 100 due to rounding error.

[24] There may, however, be variation in the population or subpopulation; we simply were unable to detect it in the sample of respondents.

APPENDIX F:

STATISTICAL PROCEDURES

STATISTICAL PROCEDURES

The descriptive comparisons discussed in this report were tested for statistical significance with a two-sided Student's t-test at the 0.05 significance level, assuming a pooled sample size of 500 or more in each comparison. With samples of this size and $p \leq 0.05$, the test will yield essentially the same results as constructing a 95 percent confidence interval around the difference between the survey estimates as discussed in Appendix E.

For multiple comparisons among groups on a *single* survey measure, a Bonferroni adjustment was applied (see Appendix E) to take into account the number of comparisons made. In this procedure, the alpha level for each individual t-test is adjusted downward so that the significance level for all comparisons among the groups will sum to $p \leq 0.05$, or whatever overall alpha level has been set for the particular group of comparisons.

As a matter of ease and economy, we assumed pooled sample sizes of 500 or more and an alpha level of 0.05 in all testing for statistical significance with the Bonferroni procedure. However, for certain comparisons, primarily those involving the American Indian/Alaska Native and Asian/Pacific Islander subgroups, the actual sample size was considerably less than 500, varying slightly in size from measure to measure depending on the number of explicit refusals, don't knows, and not sures on a particular question. Even so, the pooled sample size for any given comparison including one of those two groups was always larger than 100, with the exception of comparisons involving reasons for taking out loans (Exhibits 20 and 24).

Since the primary objective of our analysis was to explore the data for patterns rather than to confirm specific hypotheses, we chose this approach over others not only because of the economies involved, but also because it favors detection of marginally significant differences in comparisons with smaller sample sizes, precisely those comparisons where our chances of making a Type II error (not rejecting the null hypothesis of no difference, when there is a difference) are highest (see Jacob Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. [Hillsdale, N.J.: Erlbaum, 1988] for a discussion of Type II errors).

To illustrate the impact of our assumption on significance testing, Exhibit F.1 shows the adjusted p-values for the multiple comparisons and the corresponding t-values, assuming a pooled sample size of 500 or more for each comparison, and a pooled sample size of 100 for each comparison.

The first set of values in the exhibit applies to comparisons among all minority groups within the Scholar population or within the non-recipient population. It assumes all pairwise comparisons among the four minority groups are made. African Americans are compared to American Indians/Alaska Natives, to Asian/Pacific Islanders, and to Hispanic Americans. American Indians/Alaska Natives are compared to African Americans, to Asian/Pacific Islanders, and to Hispanic Americans, and so on for a total of six unique comparisons.

Exhibit F.1. Bonferroni Adjustment for Multiple Comparisons

Comparison	Total Number of Comparisons	Bonferroni Adjusted p-value for each comparison	t-value for significance testing	
			> 500 cases*	100 cases
All pairwise comparisons among minority groups within a population	6	$p \leq 0.0083$	2.65	2.69
Pairwise comparisons by minority group between two populations	4	$p \leq 0.0125$	2.51	2.54
Pairwise comparison by gender between two populations	2	$p \leq 0.025$	2.25	2.28

*used throughout this report

The second set of values in the exhibit applies to comparisons between two populations by minority group. It assumes that African Americans in one population are compared to African Americans in another population, and American Indians/Alaska Natives in one population are compared to American Indians/Alaska Natives in the other population, and so on for the other two minority groups.

The final set of values in the exhibit applies to comparisons between populations by gender. It assumes that males in one population are compared to males in another population, and females in one population are compared to females in that other population.

As the exhibit shows, the t-values for significance testing for each set of comparisons are close in value for pooled sample sizes of 500 or more and pooled sample sizes of 100. By assuming large sample sizes for all comparisons, in the first set of comparisons, we have accepted p-values between 0.0083 and 0.0094 as significant for comparisons with pooled sample sizes of at least 100 but less than 500. In the second set of comparisons, we have accepted p-values between 0.0125 and 0.0137 as significant for comparisons with pooled sample sizes of at least 100 but less than 500. Finally, in the third set of comparisons, we have accepted p-values between 0.025 and 0.0266 as significant for comparisons with pooled sample sizes of at least 100 but less than 500.

By assuming large sample sizes for comparisons on reasons for taking out loans—the only situation where the pooled sample sizes fell below 100—we have accepted p-values as high as 0.0123 as significant for comparisons between the American Indian/Alaska Native and Asian/Pacific Islander Scholars, and p-values as high as 0.0154 as significant for comparisons between the American Indian/Alaska Native Scholars and non-recipients.

APPENDIX G:

A SUMMARY OF THE SCHOLARS' BACKGROUND CHARACTERISTICS AND EARLY OUTCOMES BY MINORITY GROUP

A Summary of the Scholars' Background Characteristics and Early Outcomes by Minority Group

Who Are the African American Scholars?

- Two percent of the African American Scholars also identified themselves as American Indian/Alaska Native, 1 percent as Native Hawaiian, 1 percent as Asian, 7 percent as White, and 1 percent as ethnic Hispanic, either alone or in combination.
- Close to two-thirds were seniors in high school in one of the South Atlantic, West South Central, or East South Central states of the United States.
- Six out of every ten attended a high school where most students in their classes were not the same race they were.
- Close to one in ten attended a private high school.
- More than two out of every three took at least one advanced placement examination while in high school. About one quarter took four or more advanced placement exams in high school.
- Almost all (99 percent) took three or more years of math in high school, and 87 percent took four or more years of math while in high school.
- Almost all (98 percent) took three or more years of science in high school, and 75 percent took four or more years of science while in high school.
- All African American Scholars expect to earn at least a bachelor's degree.
- Ninety-five percent expect to go on and complete an advanced degree.
- About two out of every three expect to complete a first professional or doctoral degree.
- Most African American Scholars (99 percent) were enrolled in an institution offering at least a four-year degree.
- More than one in ten attends a historically black college or university.
- Just over half of the African American Scholars are working for pay in the current academic year.
- About one in five has taken a loan out for the current academic year.
- For about one-third of the African American Scholars, parents are contributing towards college finances.

Who Are the American Indian/Alaska Native Scholars?

- One percent of the American Indian/Alaska Native Scholars identified themselves as Native Hawaiian, 1 percent as Black/African American, 15 percent as White, and 4 percent as ethnic Hispanic, either alone or in combination.
- About four in ten belong to the Navajo or to the Cherokee Nation (Oklahoma or Unspecified).
- About four out of five were seniors in high school in one of the Mountain, West South Central, or West North Central states.
- About one in three attended a high school where most students in their classes were not the same race they were.
- Nearly one out of every five attended a private high school.
- About one out of every two took at least one advanced placement examination while in high school. Just under one in ten took four or more advanced placement exams in high school.
- Almost all (97 percent) took three or more years of math in high school, and 82 percent took four or more years of math in high school.
- Almost all (96 percent) took three or more years of science in high school, and 72 percent took four or more years of science in high school.
- Nearly all (97 percent) American Indian/Alaska Native Scholars aspire to earn at least a bachelor's degree.
- Seventy-seven percent expect to go on and complete an advanced degree.
- Nearly half expect to obtain a first professional or doctoral degree.
- Most American Indian/Alaska Native Scholars were enrolled in an institution offering at least a four-year degree.
- Four percent attend a tribal college or university.
- Over half of the American Indian/Alaska Native Scholars are working for pay in the current academic year.
- About one in ten has taken a loan out for the current academic year.
- For three in ten American Indian/Alaska Native Scholars, parents are contributing towards college finances.

Who Are the Asian/Pacific Islander Scholars?

- Four percent of the Asian/Pacific Islander Scholars identified themselves as Native Hawaiian, 1 percent as Black/African American, and 3 percent as White, either alone or in combination.
- About seven in ten have origins in Vietnam, China, or Korea.
- About three-quarters of Asian/Pacific Islanders were seniors in high school in one of the Pacific, Middle Atlantic, or South Atlantic states.
- About four out of every five attended a high school where most students in their classes were not the same race they were.
- About one in twenty attended a private high school.
- More than eight out of every ten took at least one advanced placement examination while in high school. Over one out of every two took four or more advanced placement exams.
- Almost all (99 percent) completed three or more years of math in high school, and 90 percent took four or more years of math during their high school careers.
- Almost all (99 percent) took three or more years of science in high school, and 79 percent took four or more years of science while in high school.
- All of the Asian/Pacific Islander Scholars expect to earn at least a bachelor's degree.
- Ninety-five percent expect to go on and complete an advanced degree.
- About two of every three expect to complete a first professional or doctoral degree.
- Most Asian/Pacific Islander Scholars were enrolled in an institution offering at least a four-year degree.
- Nearly four in ten are working for pay in the current academic year.
- About one in five has taken a loan out for the current academic year.
- For over four in ten Asian/Pacific Islander Scholars, parents are contributing towards college finances.

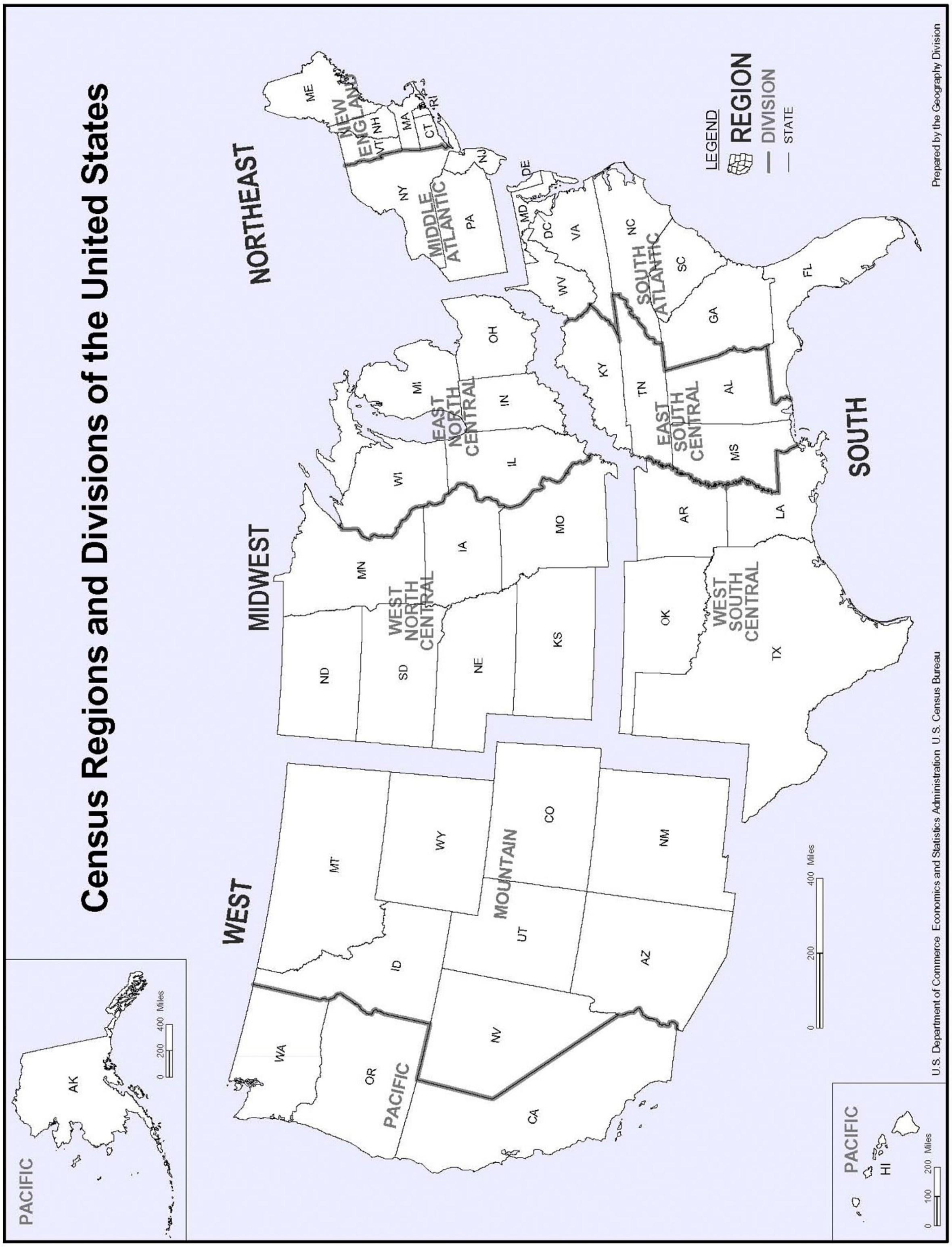
Who Are the Hispanic American Scholars?

- Ten percent of the Hispanic American Scholars identified themselves as American Indian, 3 percent as Asian, 9 percent as Black/African American, and 94 percent as White, either alone or in combination.
- About three-fourths were seniors in high school in one of the Pacific, West South Central, or South Atlantic states.
- Over half attended a high school where most students in their classes were not the same race they were.
- A little less than one out of every ten was enrolled in a religious high school.
- One out of every ten attended a private high school.
- More than four out of every five took at least one advanced placement examination while in high school. Nearly half took four or more advanced placement exams.
- Almost all (99 percent) took three or more years of math in high school, and 86 percent took four or more years of math in high school.
- Almost all (94 percent) took three or more years of science in high school, and 68 percent took four or more years of science in high school.
- All Hispanic American Scholars expect to earn at least a bachelor's degree.
- Ninety-four percent aspire to go on and complete an advanced degree.
- Close to two out of three expect to complete a first professional or doctoral degree.
- Most Hispanic American Scholars were enrolled in an institution offering at least a four-year degree.
- About half are working for pay in the current academic year.
- About one in five has taken a loan out for the current academic year.
- For nearly four in ten Hispanic American Scholars, parents are contributing towards college finances.

Appendix H:

U.S. Census Bureau Divisions of the States

Census Regions and Divisions of the United States



U.S. Census Bureau

Census Bureau Regions and Divisions with State FIPS Codes

Region I: Northeast

Division I: New England

Connecticut (09)
Maine (23)
Massachusetts (25)
New Hampshire (33)
Rhode Island (44)
Vermont (50)

Division 2: Middle Atlantic

New Jersey (34)
New York (36)
Pennsylvania (42)

Region 2: Midwest*

Division 3: East North Central

Indiana (18)
Illinois (17)
Michigan (26)
Ohio (39)
Wisconsin (55)

Division 4: West North Central

Iowa (19)
Kansas (20)
Minnesota (27)
Missouri (29)
Nebraska (31)
North Dakota (38)
South Dakota (46)

Region 3: South

Division 5: South Atlantic

Delaware (10)
District of Columbia (11)
Florida (12)
Georgia (13)
Maryland (24)
North Carolina (37)
South Carolina (45)
Virginia (51)
West Virginia (54)

Division 6: East South Central

Alabama (01)
Kentucky (21)
Mississippi (28)
Tennessee (47)

Division 7: West South Central

Arkansas (05)
Louisiana (22)
Oklahoma (40)
Texas (48)

Region 4: West

Division 8: Mountain

Arizona (04)
Colorado (08)
Idaho (16)
New Mexico (35)
Montana (30)
Utah (49)
Nevada (32)
Wyoming (56)

Division 9: Pacific

Alaska (02)
California (06)
Hawaii (15)
Oregon (41)
Washington (53)

**Prior to June 1984, the Midwest Region was designated as the North Central Region.*

Source: http://www.census.gov/geo/www/us_regdiv.pdf

NORC at the University of Chicago

APPENDIX B: DESIGN OF THE GMS LONGITUDINAL STUDY OF SCHOLARS AND NON-RECIPIENTS (NORC)

The GMS Longitudinal Study collects trend data on the impact of the scholarship program on the lives of the Scholars as they enter and progress through postsecondary institutions and into the workforce. More generally, the study will provide a wealth of information on factors that facilitate or hinder the educational and occupational attainment of students of color in the short and long term.

Designed to address the basic and applied research components of the GMS Research framework, the study follows selected cohorts of Scholars and comparison samples of non-recipients from about a year after they apply to the GMS program up until the time they reach their mid-thirties. The primary research data are collected over time through the administration of a web-based survey instrument. These data are supplemented by information from other sources to enrich the database and to provide links with national datasets.

The multiple-cohort longitudinal design of the study, the survey instruments, and supplemental sources of data are discussed below in more detail.

1. Overview of the Multiple-Cohort Longitudinal Design of the GMS Survey

1.1 Cohort Design

The multiple-cohort longitudinal design for the survey, shown below in full in Exhibit 3, includes seven distinct classes or *cohorts* of awardees and corresponding samples of non-recipients drawn from seven of the twenty GMS award years.

Exhibit 1. Cohort Design

Cohort	GMS Award Year
1	2000-2001
2	2001-2002
3	2002-2003
5	2004-2005
9	2008-2009
13	2012-2013
17	2016-2017

As summarized in Exhibit 1, the design includes four closely spaced cohorts (Cohorts 1, 2, 3, and 5)--a feature not often seen in longitudinal surveys of multiple cohorts--from the early years of the Scholarship program. Selecting multiple cohorts in the early stages of the Scholarship program benefits the research program in several ways. First, it will be possible to address a wider range of research questions and to explore certain areas in greater depth in the opening years of the program without overburdening a particular cohort with an excessively long questionnaire. Second, it will allow for the replication of early findings in a timely manner, adding strength to early conclusions about the impact of the program. Third, information on the attributes or experiences that contribute to success as a GMS Scholar or non-recipient can inform program development by the GMS partners in the early stages of the Scholarship program to enhance the overall impact of the program in the years to come. Fourth, the closely spaced cohorts will hasten the evolution of the research program in the early years of the survey, as insights from early research are incorporated into the questionnaires of cohorts that follow closely in time.

The cohort design assumes that as time goes on the focus of the research will shift to an examination of the impact of changing conditions on the educational and occupational attainment of the Scholars and non-recipients. Three additional cohorts, Cohorts 9, 13, and 17, spaced four years apart, are included in the design for this purpose.

1.2 Composition of the Cohorts

Each selected cohort consists of the full population or populations of GMS awardees and a representative sample or samples of non-recipients drawn from the same award year. In Cohort 1, there are three distinct populations of Scholars who first applied to the GMS program for support as entering freshmen, continuing undergraduates, or graduate students, and three corresponding samples of non-recipients drawn from the same award year. The remaining cohorts consist entirely of applicants who applied for support as entering. Each of those cohorts includes the full population of Scholars and a comparison sample of non-recipients from the same award year.

Thus far, the comparison samples of non-recipients have consisted of representative samples of applicants who, like the Scholars, made it past the screening phase (including the review of non-cognitive criteria by the panel members), but were disqualified or eliminated during the final verification/confirmation phase, were not asked to go on to that phase, or declined to accept the scholarship, and thus, did not receive a GMS grant. Selected to be as similar as possible to the Scholars on as many key selection variables as possible, the population of non-recipients from which the samples were drawn represent, for the most part, a segment of the population of high-achieving, low-income, minority students for whom the GMS program was designed to serve. However, without the benefit of random assignment to groups, differences between the Scholars and non-recipients remain. Designs of this type are often referred to as “non-equivalent control groups” in the literature (Cook and Campbell, 1979).

As we learn more about the impact of the GMS program, it may be worthwhile to consider the feasibility of relying on a randomized treatment design—in which qualified applicants are randomly assigned to the treatment (GMS Scholars) and control (non-recipient) conditions---in one of the remaining cohorts. Such a design would eliminate the complications associated with non-equivalent control group designs and provide stronger evidence for the impact of the program on the lives of the Scholars.

1.3 Longitudinal Design for Data Collection

The longitudinal design for data collection, summarized in Exhibit 2, allows for up to six successive rounds of data collection per cohort spanning a period of up to eighteen years per cohort.

Exhibit 2. Longitudinal Design for Data Collection

Data Collection Round	Time of Data Collection
Baseline	at the end of the applicants' first year out of high school after they have made the transition to college or the workforce
First Follow-up	at the end of the applicants' third year out of high school, which for many will correspond to their junior year in college
Second Follow-up	at the end of the applicants' fifth year out of high school, which for many will mark the transition to graduate or professional school, or into the workforce
Third Follow-up	at the end of the applicants' eighth year out of high school, which for some will correspond to the transition into the workforce after completion of graduate studies or professional school
Fourth Follow-up	at the end of the applicants' thirteenth year out of high school when most applicants will be about 31 years of age and will have completed their graduate work
Fifth Follow-up	at the end of the applicants' eighteenth year out of high school, when most applicants will be about 36 years of age

During the college years and the mid to late 20s, the design of the survey closely parallels the design of the National Longitudinal Survey of 1972, High School and Beyond, National Educational Longitudinal Study, and the Beginning Postsecondary Students Longitudinal Survey (See Appendix C), collecting information on the GMS applicants (scholars and comparison samples of non-recipients) during and after the college years (Baseline through the Third Follow-up).

The design also includes two additional rounds of data collection when the applicants are in their early and mid thirties (Fourth and Fifth follow-ups). These rounds will be especially important in capturing the long-term effects of the program on the educational attainment of Scholars who receive GMS support for graduate studies[1] and those who follow less conventional educational and occupational paths and complete their undergraduate degrees a little later in life.[2]

As shown in Exhibit 3, data collection for the survey is scheduled to end in the year 2022 with the Fifth Follow-up of Cohort 5. In other words, not all cohorts are scheduled to complete the full complement of five follow-ups. Cohorts 2, 3, 5, 9, 13, and 17 will complete their respective rounds of data collection at the time points outlined in Exhibit 2.

For Cohort 1, the pattern is a little different for a couple of reasons. First, data collection for the GMS Longitudinal Survey began in 2002. As a result, the baseline survey for entering freshmen in Cohort 1 took place at the end of their sophomore year rather than at the end of their freshmen year. Second, unlike other cohorts, Cohort 1 includes three distinct groups of Scholars who began receiving support as entering freshmen, continuing undergraduates, or graduate students. All three groups were asked to participate in the baseline survey in 2002 and in the First Follow-up in 2004 to obtain information on their early experiences in the program. After that, with the exception of the Second Follow-up for entering freshmen, the ages of the applicants at the time of data collection roughly correspond to those outlined in Exhibit 2. However, since continuing undergraduate and graduate students were older than their freshmen counterparts when they began the GMS program, fewer rounds are necessary to follow them into their mid-thirties. Data collection for graduate students is scheduled to end with a third follow-up survey that will take place when the students are about 36 years old. Data collection for undergraduate students, on the other hand, is scheduled to end with a fourth follow-up survey when the students are about 36 years old.

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[1] According to NSF's 2002 Survey of Earned Doctorates, students spend an average (median) of 7.5 years in graduate school to complete a doctorate and are 33 years old on average (median) at the time of graduation. Collecting information on the Scholars and non-recipients when they are in their thirties is thus important to capture the long-term effects of the program on the educational attainment of the Scholars.

[2] According to the National Center for Education Statistics' (NCES) 2001 Baccalaureate and Beyond Longitudinal Study of 1999/2000 bachelor degree recipients, only about half of the students who received a four-year undergraduate degree in the 1999-2000 academic year completed their degree at age 22 or younger. About 34 percent were between the ages of 23 and 29, while close to 17 percent were age 30 or older. As a whole, African, Hispanic, and Native Americans, tended to be a little older than average at the time of degree completion. Roughly 36 percent of African Americans were age 22 or younger at the time of graduation, 36 percent were between the ages of 23 and 29, and another 27 percent were age 30 or older. About 38 percent of Hispanic Americans, on the other hand, were age 22 or younger at the time of graduation, roughly 45 percent were between the ages of 23 and 29, and about 16 percent were age 30 or older. Finally, about 20 percent of Native Americans were age 22 or younger at the time of graduation, nearly 51 were between the ages of 23 and 29, with 30 percent over age 30 at the time of degree completion.

Exhibit 3. GMS Longitudinal Survey Design (NORC) Data Collection Schedule for Scholars and Non-recipients Cohorts 1, 2, 3, 5, 9, 13, 17, Years 2002 – 2011

Cohort (GMS Award Year) Nominee Groups	Data Collection Year (approximate age of nominees at the time of the survey)									
	Yr. 1- 2002	Yr.2 - 2003	Yr. 3 - 2004	Yr. 4 - 2005	Yr.5 - 2006	Yr. 6 - 2007	Yr. 7 - 2008	Yr. 8 - 2009	Yr. 9- 2010	Yr. 10- 2011
Cohort 1 (2000-2001) Freshmen (F) Undergraduates (U) Graduate Students (G)	Baseline F (20) U (21+) G (24+)		First Follow-up F (22) U (23+) G (26+)		Second Follow-up F (24)		Second Follow-up U(27+)	Second Follow-up G (31+) Third Follow-up F (27)		
Cohort 2 (2001-2002) Freshman (F)	Baseline F (19)		First Follow-up F (21)		Second Follow-up F (23)				Third Follow-up F (27)	
Cohort 3 (2002-2003) Freshman (F)		Baseline F (19)		First Follow-up F (21)		Second Follow-up F (23)				Third Follow-up F (27)
Cohort 5 (2004-2005) Freshmen (F)				Baseline F (19)		First Follow-up F (21)		Second Follow-up F (23)		
Cohort 9 (2008-2009) Freshmen (F)								Baseline F (19)		First Follow-up F (21)
Cohort 13 (2012-2013) Freshmen (F)										
Cohort 17 (2016-2017) Freshmen (F)										

GMS Longitudinal Survey Design (cont.)
Years 2012-2021

Cohort/ (GMS Award Year)/ Nominee Groups	Data Collection Year (approximate age of nominees at the time of the survey)									
	Yr.11-2012	Yr.12-2013	Yr.13-2014	Yr.14-2015	Yr.15-2016	Yr.16-2017	Yr.17-2018	Yr.18-2019	Yr.19-2020	Yr.20-2021
Cohort 1 (2000-2001) Freshmen (F) Undergraduates (U) Graduate Students (G)	Third Follow-up U (31+)	Fourth Follow-up F (31)	Third Follow-up G (36+)			Fourth Follow-up U(36+)	Fifth Follow-up F(36)			
Cohort 2 (2001-2002) Freshmen (F)			Fourth Follow-up F (31)					Fifth Follow-up F (36)		
Cohort 3 (2002-2003) Freshmen (F)				Fourth Follow-up F (31)					Fifth Follow-up F (36)	
Cohort 5 (2004-2005) Freshmen (F)		Third Follow-up F (27)				Fourth Follow-up F (31)				
Cohort 9 (2008-2009) Freshmen (F)		Second Follow-up F (23)				Third Follow-up F (27)				Fourth Follow-up F(31)
Cohort 13 (2012-2013) Freshmen (F)		Baseline F (19)		First Follow-up F (21)		Second Follow-up F (23)				Third Follow-up F (27)
Cohort 17 (2016-2017) Freshmen (F)						Baseline F (19)		First Follow-up F (21)		Second Follow-up F (23)

GMS Longitudinal Survey Design (cont.)
Year 2022

Cohort/ (GMS Award Year) Nominee Groups	Data Collection Year (approximate age of nominees at the time of the survey) Yr.21-2022
Cohort 1 (2000-2001) Freshmen (F) Undergraduates (U) Graduate Students (G)	
Cohort 2 (2001-2002) Freshmen (F)	
Cohort 3 (2002-2003) Freshmen (F)	
Cohort 5 (2004-2005) Freshmen (F)	Fifth Follow-up F (36)
Cohort 9 (2008-2009) Freshmen (F)	
Cohort 13 (2012-2013) Freshmen (F)	
Cohort 17 (2016-2017) Freshmen (F)	

APPENDIX B: DESIGN OF THE GMS LONGITUDINAL STUDY OF SCHOLARS AND NON-RECIPIENTS (NORC)

2. Overview of the Design of the Survey Instrument

The survey instrument for the study includes several core components of questions that will be asked of every cohort in one or more rounds of data collection to allow for cross-cohort comparisons and the evaluation of trends within and across cohorts. Some of these questions will continue to be drawn from other major longitudinal studies of educational and career attainment to allow for critical comparisons with prominent national surveys.

As the research evolves, the questionnaire for some cohorts may include cohort-specific modules and constructs to explore an area of research in greater depth. In other words, the cohorts may have their own more focused research agenda that goes beyond what can be addressed with the core components of the questionnaire. With cohort-specific modules, it will be possible to address a wider range of research questions in greater depth without overburdening the members of any given cohort.

In many cases, addressing the research objectives will require information in addition to that collected with the questionnaire. In the next section, we discuss the plan for adding or linking supplemental information to the GMS Longitudinal databases.

3. Sources of Additional Information for the GMS Longitudinal Databases

As part of the study, data from the web-based survey will be supplemented with information from other sources to support the research goals of the GMS framework. Some information like the Integrated Postsecondary Database Systems Codes (IPEDS Codes) and census tract codes for postal zip code at the time of high school graduation will be added to the databases of all cohorts. Other information will vary by cohort depending on its availability, the priorities of the Foundation, the evolution of the research framework, and the research agenda of the particular cohort.

Exhibit 4 summarizes the current plan for adding and linking information to the GMS databases. The first column in the exhibit lists the source of information, the second column provides a brief description of the information, and the remaining columns, one for each cohort, indicate whether the information is scheduled to be added or linked to the database of a particular cohort under the current plan.

To date, there are eight supplemental sources of information that will be added or linked to one or more of the databases. As indicated in Exhibit 5 by the shaded cells, the first three sources of information, the Integrated Postsecondary Database System codes (IPEDS codes), the Census Tract codes, and the Industry and Occupational Codes (IO codes) will be added to the databases of all cohorts. IPEDS codes are generated for all past and current postsecondary institutions of enrollment, as well as the applicant's first, second, and third choice of institution. New codes are added to each round of data collection as Scholars and non-recipients change their postsecondary institutions of enrollment. Census Tract codes, on the other hand, are generated for the applicant's postal zip code at the time they graduated from high school and remain the same throughout the study. Like the IPEDS codes, new IO codes are added with each round of data collection to reflect the applicant's current job and occupational aspirations.

The IPEDS codes provide for links with the Integrated Postsecondary Database System and other databases that rely on these codes. The system includes aggregate information on postsecondary institutions of education in such areas as educational offerings, enrollment, graduation rates, and finances. The Census Tract codes provide for links with the U.S. Census data. In addition to adding the codes to the databases, NORC will extract the actual census tract data and store it in a separate database to allow for easy access to the information.

The fourth source of information, the NCES High School Codes for the applicants' high school of graduation, was kindly supplied by Bill Trent, one of the RAC members, for the members of Cohort 2. These codes allow for links with NCES' Common Core of Data (CCD) and other databases that rely on them. The CCD includes annual school-level information on all public high schools in the United States.

Exhibit 4. Sources of Additional Information for GMS Longitudinal Databases*

Source	Description	Cohort						
		1	2	3	5	10	15	20
Integrated Postsecondary Database System (IPEDS) Codes	IPEDS codes for the postsecondary institution of past and current enrollment, and the first, second, and third institutions of choice as reported by the applicant. These codes provide links to the NCES' IPEDS of institution-level data in such areas as educational offerings, enrollment, graduation rates, finances, faculty, and so on.							
Census Tract Codes {geocodes} and Information	Census tract codes for the zip code of the applicant at the time of high school graduation. These codes provide for links with census tract information, which is also extracted.							
Industry and Occupational Codes (IO codes)	IO codes for the applicants' current job if working off campus and occupational aspirations, both at the time of the respective round of data collection.							
NCES High School Codes	NCES code for the applicant's high school of graduation. Provides for links with NCES' Common Core of Data (CCD) and other datasets that use these codes. The CCD includes annual school-level information on all elementary and secondary public schools in the U.S.							
U.S. News and World Report Rankings	Rankings from the U.S. News and World Report along with information that was used to create a selectivity index for the colleges							
High School Transcript Information	Transcripts of courses taken during high school along with other background information. This information must be keyed and coded before it can be added to the databases.							
Non-cognitive variables and composite scores	Selection criteria from the GMS application process, reader ratings, leadership ability, community involvement, self-concept, long-range goals and so on.							
Data Warehouse (UNCF) (under construction)	Information from the GMS application and renewal processes.	Added as needed when the warehouse becomes accessible						

*Shaded areas indicate that the information will be added or linked to the database of the specified cohort or cohorts

The fifth source of information, the U.S. News and World Report Rankings, was again supplied by Bill Trent for the members of Cohort 2. It includes rankings for the various colleges, their IPEDS codes for linking the information to the IPEDS codes in the GMS databases, along with information used to create the U.S. News and World Report selectivity index.

The sixth source of supplemental data, high school transcripts, includes information on course enrollment, credits, grades, class rank, overall and yearly GPA, standardized test scores and honors, and disability status. UNCF collects high school transcripts as part of the GMS application process.

From Cohort 4 on, authorization to use the information in the applicants' high school transcripts for research purposes is automatically secured as part of the GMS application process.[3] This was not the case in Cohorts 1, 2, and 3. NORC is currently in the process of securing authorization from those applicants as part of the first follow-up of Cohorts 1 and 2, and the tracking of Cohort 3. Depending on the rates of authorization, NORC may consider subsampling refusals and non-respondents and following up with those individuals to boost the authorization rates and representativeness of the subsamples of applicants granting permission.

As indicated in Exhibit 5, the current plan provides for coding of the high school transcripts from Cohorts 1 and 2. Coding of Cohort 3 transcripts is under consideration.

From the coding, it will be possible to generate measures of academic intensity and quality (Adelman, 1999) and other indicators of academic preparation to assess the contribution of high school preparation on the success of the GMS applicants, as Scholars and non-recipients.

The seventh source of information consists of the non-cognitive variables employed as part of the GMS selection process. As indicated in Exhibit 5, NORC will add these variables to the databases of all cohorts. Among others, they include indicators of self-concept, community involvement, leadership, nontraditional knowledge acquired, and long-range goals.

The eighth and final source of information is the UNCF data warehouse, currently under construction. It will include information from the application and renewal process of all cohorts. NORC will add information from this source to the GMS databases of all cohorts as it becomes available and needed.

[3] Transcript information will thus be available for all Scholars and non-recipients from Cohort 4 onward.

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